

Sub Code: CST022 ROLL NO.....

EVEN SEMESTER EXAMINATION, 2024 – 25

Third Year (VI Sem) B.Tech.: CS&E

ARTIFICIAL INTELLIGENCE

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) What is the difference between informed and uninformed search strategies in AI? (ii) Discuss constraint satisfaction problem in AI with suitable example. b) Discuss the evolution of Artificial Intelligence from its early history to the current state-of-the-art technologies. Include key milestones, challenges, and major breakthroughs. c) Explain the concept of a problem-solving agent in AI. Describe how such agents operate using uninformed and informed search strategies, and provide examples of each.
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) What are the main differences between propositional logic and predicate logic in knowledge representation? (ii) Explain the concept of unification in predicate logic with an example. b) Describe the various approaches to knowledge representation in AI. Discuss the major issues associated with these approaches, such as consistency, completeness, and inferencing efficiency. c) Differentiate between weak slot-filler structures and strong slot-filler structures. How are they used in building knowledge-based agents, and what are their advantages and limitations?

Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) State Bayes' Theorem and explain its significance in AI reasoning under uncertainty. (ii) What is the role of conditional probability in Bayesian networks? b) Explain the structure and components of a Bayesian Network. How is it constructed, and how can inference be performed using it? Provide a suitable example. c) Compare and contrast Neural Networks, Fuzzy Logic, and Genetic Algorithms in terms of their approach to solving AI problems. Highlight their strengths, limitations, and typical application areas.
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) What is conditional planning, and how does it differ from classical state space planning? (ii) Briefly explain the concept of reinforcement learning with a real-world example. b) Discuss different types of planning strategies in AI, including state space search, conditional planning, continuous planning, and multi-agent planning. How do these approaches handle dynamic environments and uncertainty? c) Explain and compare the different forms of learning in AI: inductive learning, statistical learning,

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262

Even Semester Examination 2017-18

B.Tech. (SEMESTER-VI)

ARTIFICIAL INTELLIGENCE

Time: 03:00 Hours

Note: (i) Attempt ALL question. All Question carry equal marks.

Max Marks: 100

1.

Attempt **any four** parts of following:

(5x4=20)

(a)

What is AI? Explain any five applications area of AI in short.

(b)

Explain the method of Hill climbing. Also explain the problems associated with hill climbing and possible solutions.

(c)

Explain A* Algorithm by an example.

(d)

Discuss AO* algorithm. Give one example where AO* is suitable to apply.

(e)

(f)

Define "Heuristic Search". Explain the steps in "Best First Search" and illustrate it using a suitable example.

Solve following cryptarithmic problem with appropriate strategy/steps:

EAT

+ THAT

APPLE

Attempt **any four** parts of following:

2.

(a)

(5x4=20)

What are the limitations of Propositional Logic? Explain how they can be

overcome using Predicate logic.

(b)

Explain knowledge based agent with example.

TCS-603/1800

(1)

[P.T.O.]

(c)

Explain unification in predicate logic. Also discuss the steps of converting predicate logic wffs to clause form.

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(d)

Explain Semantic Net & Frame with suitable example.

(e)

Explain Strong slot filler with example?

(f)

Consider the following sentences: I JADRITRA

001:2h6M

Tennis is a game. Chess is a game.

(ii)

John and Steve are students.

(iii)

John plays Tennis.

(iv)

Steve plays everything that John plays.

(v)

Students who play Tennis, do not play Chess.

20H 00:50 29mT

nd diw botsia

Translate the above sentences into formulas in Predicate logic and Prove using resolution that "Steve does not play Chess"

3.

Attempt **any** two parts of following:

(a)

(10x2=20)

What is non-monotonic reasoning? How AI handles reasoning under uncertainty?

Explain with example

State the Baye's theorem. Explain Bayesian Network in detail.

(b)

(c)

Write short notes on

(i)

Dempster Shafer Theory

(ii)

Fuzzy Logic

4.

Attempt **any two** parts of following:

(a)

(10x2=20)

Explain the concept of planning with state space search with an example. Explain conditional planning.

(b) Explain learning decision trees with an example.

TCS-603/1800

(2)

(c)

Write short notes on:

(i)

Neural Net learning

(ii)

Genetic learning

5.

Attempt any **two** parts of following:

(a) Write short notes on

(i)

Alpha - beta pruning

(ii)

Knowledge acquisition stages.

(10x2=20)

(b) What is the swarm intelligent system? Explain the application and working of Ant Colony System.

(c)

Define expert system. Also explain MYCIN expert system.

TCS-603/1800

(3)

PAPER ID-

EXO TCS 603

Roll No. to be filled in your Answer Book

Roll No.

Time- 3 Hours

NOTE:

B.Tech

VI

room to **mCSE, VI**

Artificial Intelligence (TCS 603)

i.

All questions are compulsory.

ii.

Draw diagrams wherever necessary.

iii.

All questions carry equal marks..

1. Attempt any **FOUR** parts of the following.

(A)

Max marks: 100

5X4

(B)

(C)

What is the difference between strong artificial and weak artificial intelligence.

Distinguish between forward and backward chaining?

What is the common way to represent and parse Grammars for natural language processing?

(D)

Explain state space approach for solving any AI problem.

(E)

What are the differences and similarities between problem solving and planning?

(F)

What is Artificial Intelligence? Explain with example.

2. Attempt any **FOUR** parts of the following

(A) What is A* algorithm search method?

(B) Explain breadth first search algorithm?

Explain propositional and predicate logic?

What is fuzzy logic? List all the applications of fuzzy logic.

(C)

(D)

(E)

Explain multi-agent planning with the help of example?

(F)

Write short note on game playing.

5 X 4

3. Attempt any **TWO** parts of the following

10X2

(A) Discuss the role of reasoning in AI. How predicate logic is used in AI to represent

(B)

(C)

knowledge?

Explain A* searching technique in detail with example. Discuss conditions for the optimality of this technique.

Explain back propagation algorithm for neural nets learning?

Spillstal

4. Attempt any **TWO** parts of the following

10X2

(A) Explain all the issues in knowledge representation?

(B)

Explain the following expert system:

a) MYCIN

b) DART

(C)

Explain iterative deepening depth first search algorithm?

5. Attempt any **TWO** parts of the following

10X 2

bop(A) in What is difference between inductive, deductive, abductive learning?

(B) What is expert system and characteristics of expert system with example? orgnuging

(C) What is Ant-Colony System? Explain its application and working.

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TCS 603

No. of Printed page: 02

Paper Code TCS-501

76

Paper ID and Roll No. to be filled in your Answer book

Roll. No.

B. Tech.

Computer Graphics

[Time: 3Hours]

Q1.Attempt any four: -

Total Marks: 100

(4X5=20)

- a. What are the different components of modern graphics display?
- b. what is computer graphics? Explain the brief various application areas of computer graphics.
- c. Explain the role of pixel and frame buffer in graphics devices.
- d. Define the graphics primitives. Mention some typical graphics primitives. e. What is CRT? What are problems with the CRT as a computer display? f. Describe the character generation in computer graphics.

Q2 Attempt any four :-

- a. Discuss the Bresenham's line algorithm with an example.

(4X5=20)

- b. Consider three different raster systems with resolution of 640 x 480, 1280 x 1024, 2560 x 2048. What size of frame buffer (in bytes) is needed for each of these systems to store 12 bits/pixel?
- c. Explain the polygon representation. What is the use of polygon in computer graphics?

- d. Explain the display **file** structure with suitable example.
- e Draw a line from point (-1,-1) to point (4,5) using the bresenham's **algorithm**.
- f. Explain the **scan** line and boundary fill algorithm with example.

Q3. Attempt any two: -

08-2014 **(10X2=20)**

a) Use **the Cohen** sutherland algo **to** clip **the line** P1 (70, 20) and P2 (100, 10) against a window **lower** left hand corner (50, 10) and upper right hand corner (80, 40).

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b) consider the square A(1,0),B(0,0),C(0,1),D(1,1). Rotate the square ABCD by 45° clockwise about A (1, 0).

c) Explain **the** rotation transformation and translation transformation with example.

Q4. Attempt any two

(10X2=20)

a) Explain the 3D transformation reflection about X-axis and Y-axis in detail.

b) What do you mean by parallel projection explain with suitable example.

c) Write a short notes on

Axonometric Projections

Oblique Projections

3D Transformation

Q5. Attempt any two :-

(10X2=20)

a) Define The Sierpinski Gasket and its applications.

b) Write a short notes on:

The OpenGL API

Three-Dimensional Gasket

c) Explain the OpenGL API primitives, attributes and controlling functions.

TCS-501/1700/2

TCS-501

1347

Odd Semester Examination, 2017-18

B.TECH. (SEMESTER-V) sed

COMPUTER GRAPHICS

Time: 03:00 Hours

Note: Attempt **all** questions.

1.

Attempt **any four** questions:

(a)

Max Marks: 100

(5x4=20)

Consider the line coordinates (0,0) and (8,4), Rasterize the line segment using Bresenham's algorithm.

(b)

Explain scanline fill algorithm with an example showing necessary vertex list, edge list, active edge list etc.?

(c)

Justify the approach of using integer arithmetic in Bresenham line drawing algorithm. Explain how rasterization accuracy is preserved despite using integer arithmetic?

(d)

Draw the flow chart for midpoint circle generation algorithm.?

(e)

Explain the basic concept of midpoint ellipse drawing algorithm. Derive the decision parameter for the algorithm ?

Attempt **any four** questions:

2.

(a)

(5x4=20)

Explain difference between polygon relative and polygon absolute algorithm of entering polygons?

(b)

Explain about the painter's algorithm. Explain the situation where the painter's algorithm does not work properly.

(c)

What is segment? How is segment useful to draw and redraw a part of image and how is segment table created?

(d)

TCS-501/1600

Derive the window to view port transformations equations by first scaling the window to the size of the viewport and then translating the scaled window to the view port position.

(1)

[P.T.O.]

(e)

Write Short notes on:

(i)

Double Buffering

(ii)

Diffuse reflection

(iii)

Dot Matrix Method

Attempt **any two** questions:

3.

(a)

(10x2=20)

Compare the orthographic and oblique types of parallel projections and also explain the various clipping parameters in 3D clipping?

(b)

Using homogeneous coordinate transformation matrix, apply the following sequence of transformation to a unit square centered at origin:

(i)

Translation by factor $(1/2, 1)$

(ii)

Rotate by an angle $= 90^\circ$

(c)

Explain the Cohen Sutherland Line Clipping Algorithm? also Use this algorithm to find the visible portion of the line $P(40, 80)$ $Q(120, 30)$ inside the window, the window is defined as ABCD: $A(20,20)$, $B(60, 20)$, $C(60, 40)$ AND $D(20, 40)$.

Attempt **any two** questions:

4.

(a)

(i)

(10x2=20)

Derive the 2×2 transformation matrix for each of the following rotation about the origin, counterclockwise by π and clockwise by $\pi/2$.

(b)

(ii)

Consider a graphics **system** which allows the user to have more than one segment such a system. What would be its advantages and disadvantages?

Obtain a two point perspective transformation of a unit cube situated at the origin with center of projection at $x = -2$ and $y = -2$ projected onto the $z = 0$ plane.

(c)

Explain :

(i)

Plain Projection and its Types

(ii)

Oblique Projections

(iii)

Z-Buffer Method and its disadvantages

5.

Attempt **any two** questions:

(10x2=20)

(a) Write Short Notes on Open GL API and Sierpinski Gasket?

(b)

List the properties of Animating Interactive Programs? Describe 3-D projection clipping procedure in detail?

TCS-501/1600

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(2)

Paper Code -TCS502

12

No. of Printed page:2

Paper ID and Roll No. to be filled in your Answer book

Roll. No.

[Time: 3Hours]

B. Tech.

(SEM. VI) DEC-2014 **Computer Network**

Total Marks: 100.

Note: Attempt all question

Q:-1 Attempt any 4 questions

(a) What are the responsibilities of session layer?

5X4=20

(b) What is the difference between service point address, logical address and physical address?

(c) How do the layers of TCP/IP protocol suite correlate to the layers of OSI?

(d) Discuss LRC and the type of errors it can and can't detect.

(e) What is the difference between point to point and multipoint line configuration.

(f) Explain the term networking and internetworking. Write application of Computer network.

Q:-2 Attempt any 4 questions

(a) Explain in detail the TCP/IP reference model.

5X4=20

(b) What do you mean by the term "Quality of Service (QoS)".

(c) What are the various design issues for the layers?

(d) What is the remainder obtained by dividing $X^3 + X + 1$ by the generator polynomial $X^3 + 1$?

(e) Sketch the Manchester encoding for the bits stream

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Q:-3 Attempt any 2 questions

10X2=20

(a) Give two examples application for which connection **oriented** service is appropriate and connectionless service is best.

(b) An ATM network uses a token bucket scheme for traffic shaping, a new token is put in to the bucket every $5\mu\text{sec}$. what is the maximum sustainable netdata rate (i.e excluding header bit).

(c) A large FDDI ring has 100 stations & a token rotation time of 40 msec. the token holding time is 10 msec. What is the maximum achievable efficiency of the ring?

Q:-4 Attempt any 2 questions

10X2=20

(a) What do you mean by traffic shaping. Explain only one of **the** technique by which traffic can be shaped.

(b) Explain Open loop congestion control and closed loop congestion control in detail.

(c) What is the difference between non persistent CSMA/CD & p persistent CSMA/CD.

Q:-5. Attempt any 2 questions

10X2=20

(a) Explain IPV6 with suitable diagram. Is it possible to transmit data from IPV6 to IPV4? Justify your answer.

(b) Consider building a CSMA/CD network running at 1 gbps over a 1 km cable with no repeaters. The signal speed in the cable is 200,000 km/sec. what is **the** minimum frame size.

(c) Write short notes on the following:

1. Frame Relay

TOTE 2. ISDN

3. Slotted ALOHA

TCS-502/1830/2

Sub Code: BCST602 ROLL NO.....

EVEN SEMESTER EXAMINATION, 2023 – 24

3rd yr B.Tech. – Computer Science & Engineering

Compiler Design

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) What is difference between Compiler and Interpreter? b) Discuss the action taken by every phase of compiler for the following statement. Position:= initial+rate*60. c) Design NFA by using Thompson's construction algorithm for regular expression (a b)*abb. d) Construct predictive parsing table for the grammar S→iEtSeS iEtS a, E→ b. e) Discuss Operator Precedence Parsing algorithm with example. f) What is ambiguous grammar? identify whether following grammar is ambiguous or not. S→ aSbS bSaS λ.	5x4=20
Q 2.	Answer any four parts of the following. a) What are the various storage management techniques in symbol table? b) Draw the syntax tree and DAG for the following expression: (a*b)+(c-d)*(a*b)+b c) What is activation record? Write the various fields of Activation Record. d) Classify the errors and discuss the errors in each phase of Compiler. e) What are basic blocks? Write the algorithm for partitioning into Blocks. f) What is control and data flow analysis? Explain with example.	5x4=20

Q 3.	Answer any two parts of the following. a) Show the following grammar $S \rightarrow Aa bAc Bc bBa$, $A \rightarrow d$, $B \rightarrow d$ Is LR(1) but not LALR(1) b) Write the properties of LR parser with its structure. Also explain the techniques of LR parser. c) What is LALR parser? Construct the set of LR(1) items for this grammar: $S \rightarrow CC$, $C \rightarrow aC$, $C \rightarrow d$	10x2= 20
Q 4.	Answer any two parts of the following. a) Write quadruples, triples and indirect triples for the expression: - $(a*b)+(c+d)-(a+b+c+d)$ b) Write the short note on: i. Abstract syntax tree ii. Polish notation iii. Three address code iv. Backpatching c) Write the definition of symbol table and procedure to store the names in symbol table.	10x2= 20
Q 5.	Answer any two parts of the following. a) Write the comparison among Static allocation, Stack allocation and Heap Allocation with their merits and limitations b) What is code optimization? Explain machine dependent and independent code optimization.	10x2= 20

c) Consider the following program code: <pre> Prod=0; l=1; Do{ Prod=prod+a[i]*b[i]; l=i+1; }while (i<=10); i. Partition it into blocks </pre>	
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	ii. Construct the flow graph	
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Sub Code: CST-012 ROLL NO..... ODD SEMESTER EXAMINATION,
2024 – 25

III Year (V Sem) B.Tech.: Computer Science & Engineering

COMPILER DESIGN

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) Describe the languages denoted by the regular expressions: $a^*ba^*ba^*ba^*$. (5 marks) (ii) Explain the phases of compiler in brief. (5 marks) b) Consider the context-free grammar $S \rightarrow SS^+ SS^*a$ Show how the string $aa+a^*$ can be generated by this grammar. Construct a parse tree for this string. What language does this grammar generate? Justify your answer. (10 marks) c) Construct the minimum state DFA's for the regular expression $(a b)^*a(a b)$. (10 marks)
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) Explain left-factoring left recursion. (5 marks) (ii) Define YACC. Outline steps required to create an input/output translator with YACC. (5 marks) b) Show that the following grammar: (10 marks) $S \rightarrow AaAb BbBa$ $A \rightarrow \epsilon, B \rightarrow \epsilon$. is LL(1) but not SLR(1). c) Parse the input string aab using LR(0) on given grammar $S \rightarrow AA$ $A \rightarrow aA$ $A \rightarrow b$ (10 marks)

Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) Write down translation procedure for control statement. (5 marks) (ii) Differentiate between L-attributes and S- attributes with example. (5 marks) b) Define quadruples, triples and indirect triples. Write quadruples, triples and indirect triples for the expression: $(a + b) * (c + d) - (a + b + c)$. (10 marks) c) Define syntax directed translation. Construct annotated parse tree for the expression $(4 * 7 + 1) * 2$. How the semantic action attached to production? (10 marks)
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) Explain activation record and its contents with diagram. (5 marks) (ii) Define heap management. Explain memory manger functions. (5 marks) b) Explain garbage collection. Point out performance metric that must be consider to design a garbage collector. Also discuss reachability. (10 marks) c) Give the DAG representation of basic blocks and perform code improving transformations on the code represented by the block. Also, explain peephole optimization. (10 marks)

Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) Write sources of redundancy and partially redundant expression. (5 marks) (ii) Describe the lazy code motion problem. (5 marks) b) Explain what constitute a loop in flow graph and how will you do loop optimization in code optimization of a compiler. (10 marks) c) Describe data flow analysis. Explain how does it used in code optimization. (10 marks)
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TCS-602

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Even Semester Examination, 2017-18

B.TECH. (SEMESTER-VI)

COMPILER DESIGN

E

Time: 03:00 Hours

Max Marks: 100

Note: Attempt **all** questions.

Q1.

Attempt **any four** Questions:

(4 × 5 = 20)

F

(a) What are the various phases of the compiler. Explain each phase in detail.

(b)

Write down the various BNF notations and Explain the significance of each in detail.

What is LEX and YACC. What is the format of LEX input file.

(c)

(d)

What is the role of lexical analyzer. Explain the process of constructing the NFA from regular expression.

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(e)

Why are multiple passes required in a compiler. Describe strategies for reducing the number of passes.

(f)

Describe Boot strapping and its uses in compiler design.

22. Attempt **any four** questions($4 \times 5 = 20$)

(4

(a) Check whether the following grammar is LL(1) grammar

$S \rightarrow E \mid S E \mid a$

$E \rightarrow b$

(b)

-->

What do you understand by ambiguity in grammar? How the grammar is made unambiguous using precedence order and associativity among arithmetic operators.

TCS-602/1800

(1)

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[P.T.O.]

(c) Compute first and Follow and construct the predictive parsing Table for the following grammar

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

(d)

001

(e)

(1)

$F(E) \mid id$

---->

Write the quadruple, Triple and indirect triples for the following equation:

$$(X+Y) * (Y+Z) + (X+Y+Z)$$

What language is generated by the following grammar?

$$s \Rightarrow 0s1 \mid 01$$

Construct minimum state DFA for the following regular expression

$$(alb)^*a(alb)$$

Attempt **any two** Questions:

Q3.

(a)

Consider the following grammar:

$$(2X \ 10 = 20)$$

unitbuntenos ABC

$$A \Rightarrow a \mid bbD$$

$$Ba \in$$

$$C \rightarrow b \in$$

$$D \Rightarrow c \in$$

tos = Construct the first and follow sets for the grammar and also design a LL (1)

(b)

parsing table for the grammar.

Consider the grammar -

$$S \ ACB/C \ b \ B/B \ a$$

$$A \ da \ /BC$$

$$B \ g/\in$$

Ch/€

Calculate FIRST, FOLLOW and Construct the predictive Parsing Table.

TCS-602/1800

(2)

(c) Construct the LALR parsing table for the following grammar

$S \rightarrow AA$

$A \rightarrow aA$

A

-->

-->

b

(i)

Generate the canonical sets of LR (1) items.

(ii)

Construct the LALR parsing table.

Q4.

Attempt **any two** Questions:

(2 × 10 = 20)

(a)

Explain the logical Phase Error and Syntactic phase error. Also suggest methods for recovery of errors.

(b)

What is Symbol Table? Explain its capabilities, also Mention the name suitable data structure for symbol table and explain it in detail.

(c)

Consider the following grammar and give the syntax directed definitions to

construct parse tree. For the input expression $4*7+1*2$ construct an annotated parse tree according to your syntax directed definition:

$E \rightarrow E*T \mid T \mid +T$

$T \rightarrow T/F \mid F$

$F \rightarrow \text{digit}$

Q. 5. Attempt **any two** Questions:

(a)

(b)

$(2 \times 10 = 20)$

Describe the various storage management techniques in detail. Also describe DAG and explain how it is useful in the code optimization.

Give the syntax directed translation scheme to translate the while control construct. Also translate the following program segment into three address code:

While($a > b$)

If($c > d$)

TCS-602/1800

(3)

[P.T.O.]

$c = c - d * e;$

else

$c = c + d * e;$

(c)

Write short notes on the following:

(i)

Local and loop optimization

A

(ii)

Induction variable elimination

TCS-602

1091

Even Semester Examination 2018-19

B.Tech. (Computer Science Engg.) (SEMESTER-VI)

COMPILER DESIGN

Time: 03:00 Hours

Max Marks :100

SECTION-A

(Attempt **any four**. All question **carry equal** marks)

[4x5 = (20 marks)]

Q1. Define the following terms:

a.)

Phases and passes

b.)

BNF notation

c.)

Operator Precedence

d.)

Error Detection & Recovery

e.)

Loop optimization

SECTION-B

(Attempt **any four**. All question **carry equal** marks)

[4x5= (20 marks)]

1.

Describe the tasks that are performed by semantic analysis phase of compiler.

2.

Find the FIRST and FOLLOW function for the following grammar:

$S1=R$

$S \Rightarrow R$

L^*R

Lid

R L

TCS-602/1840

(1)

[P.T.O.]

3.

3.

What do you mean by Looping unrolling? Explain its importance.

4.

Explain Finite state machines and regular expressions with the help of one example.

5.

Explain with the help of diagram the Global Data-Flow analysis.

SECTION-C

(Attempt **any four**. All question **carry equal** marks)

[4×5 = (20 marks)]

1.

Explain the concept of input buffering with sentinel in lexical analysis.

2.

Construct SLR parsing table for the following grammar: $SSS +/SS^* / a$.

3.

Explain the context free grammar with suitable example.

4.

Differentiate the syntactic phase errors and semantic errors.

5.

Define the code optimization with the help of any example.

SECTION-D

(Attempt **any two**. All question **carry equal** marks)

1.

[2×10 (20 marks)]

Write the algorithm for finding the canonical collection of LR(O) items. Define the closure (1) and goto (1, x) functions.

2.

Check whether the following grammar is ambiguous or not. If it is ambiguous, remove the ambiguity from the grammar:

$E \rightarrow$

$E+T / T.$

$T \rightarrow T * F / F$

$F \rightarrow (E) / id$

Construct NFA to DFA for the regular j expression $(a/b)^*ab^*a$. First show that constrictions of NFA then demonstrate the DFA construction.

TCS-602/1840

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(2)

SECTION-E

(Attempt **any two**. All question **carry equal** marks)

1.

2.

[2×10= (20 marks)]

What is Translator? Discuss the role of various phases of complier in translation of source code to target code.

Explain the importance of intermediate code generation phase in phases complier.

What are the different issues that must be considered in the selection of intermediate codes?

3.

Explain the DAG representation of basic blocks, value numbers and algebraic laws with the help of suitable example.

TCS-602/1840

(3)

X

Time: 03:00 Hours

1205

Odd Semester Examination 2018-19

B.TECH. (CSE) (SEMESTER-V)

COMPUTER GRAPHICS

Section A

Note: Attempt **any four**. All question carry equal marks:

1.

Define the following terms:

(a)

Convex hull

(b)

HIS Model

SB (c)

Polygon Meshes

(d)

Cohen-Sutherland Line Clipping

(e) Illumination Model

Section B

Note: Attempt **any four**. All question carry equal marks:

Max Marks: 100

[4x5=20]

[4x5=20]

1.

Differentiate between Bezier and B-spline curve.

2.

Construct a line using DDA algorithm in which initial point is (0,0) and final point is (6,7).

3.

What do you mean by reflection and shearing, explain with diagram and matrix form.

4.

Differentiate between beam-penetration and shadow mask method using suitable diagram.

What steps are required to scan convert a circle using Bresenham's algorithm.

5.

TCS-501/2120

(1)

[P.T.O.]

Section C

Note: Attempt **any four**. All question carry equal marks:

[4x5=20]

1. State and explain the weileratherton polygon clipping algorithm with proper diagram.

2.

3.

4.

5.

Translate the pyramid defined by the co-ordinates A(0,0,0) B(1,0,0) C(0,1,0)

D(0,0,1), 2units each in x,y,z directions.

use the cohen-sutherland algorithm to find the visible portion of the line P(40,80) and Q(120,30) inside the window the window is defined as ABCD: A(20,20), B(60,20), C(60,40), D(20,40).

Differentiate between raster scan and random scan systems.

What steps are required to generate a circle using the polynomial method

Section D

Note: Attempt **any two**. All question carry equal marks:

[2×10=20]

1.

Obtain the mirror reflection of a triangle formed by vertices A(0,3) B(2,0) and C(3,2) about line passing through the points (1,3) and (-1,-1).

2.

Give a single 3*3 homogenous coordinate transformation matrix, which will have same effect as each of the following transformation sequences. [10]

(a)

Scale the image to be twice as large & then transmit it 1 unit to left.

(b)

Scale the X direction to be one-half as large & then rotate counter clockwise by 90 degree about the origin.

(c)

Rotate counter clockwise about the origin by 90 degree and then scale the X direction to be one -half as large

(d)

Translate down one-half unit, right one-half unit & then rotate counter clockwise by 45 degree.

TCS-501/2120

(2)

3.

State Sutherland-hodgeman polygon clipping algorithm and also mention its advantages and state the differences between concave & convex polygon.

Section E

Note: Attempt **any two**. All question carry equal marks:

1.

[2×10=20]

Consider following window coordinates A(100,10) B(160,10) C(160,40) and D(100,40). Find visible portion of line segments EF, GH and IJ using cohen Sutherland

algorithm. The points are E(50,0), F(70,80), G(120,20), H(140,80), I (120,5) and J (180,30).

2.

3.

Draw line with the help of Bresenham's algorithm for following given points P1(20,10) & P2(28,22).

Write down the differences between beam-penetration and shadow mask method explain it with the help of suitable diagram And how much time is spent scanning across each row of pixels during screen refresh on a raster system with a resolution of 640*480 and a refresh rate of 60 frames per second.

TCS-501/2120

(3)

Roll no.

Total no. of printed pages: 2

ODD SEMESTER EXAMINATION, 2019-20 B. Tech (CSE, 5th Semester) COMPUTER GRAPHICS

Time: 3 hour

Max. Marks: 100

Note: (i)

*Attempt **ALL** questions.*

(ii)

All Questions carry equal marks.

(iii)

Be precise in your answer.

1.

Attempt **any four** parts of the following:

(5 x 4 = 20)

(a)

Explain how to display file structure and control test?

(b)

Explain in detail DDA algorithm of line?

(c)

A laser printer is capable of printing two pages (**size** 9*11 inch) per second at resolution of 900 pixels per inch. How many bits per second does such device require?

(d)

Explain the different display devices?

(e)

Explain the display file structure.

(f)

Explain what do you mean by OpenGL API?

2.

Attempt **any four** parts of **the** following;

(5×4=20)

(a)

Explain the different polygon filling algorithms.

(b)

(c)

Write procedures for creation and deletion of segment with suitable examples?

What is a segment and segment table? Write the utility of segment.

(d)

Explain visibility in detail.

(e)

Explain the concept of fractured segment with suitable diagram if required. You are also required to explain the structure of display file in brief.

(f)

Define windowing and viewport. Derive window to viewport transformation.

1

PTO

3.

(a)

(b)

Attempt **any two** parts of the following:

(10 × 2 = 20)

Describe the Cohen Sutherland polygon algorithm with suitable example.

Explain Liang barsky line clipping algorithm..

(c)

state and prove the following 2d transformations matrix

4.

1. Rotation

2. scaling

3. shearing

Attempt **any two** parts of the following:

(a)

(10 × 2 = 20)

What are various back face detection algorithm? Explain any one of them.

Determine transformation matrix for rotating (clockwise or anticlockwise) a point in 3-D around **the** line $y=x$ in x-y plane by angle .

(b)

(c)

With suitable examples, explain all 3D transformations.

Attempt **any two** parts of the following:

(10 × 2 = 20)

Briefly explain Sierpinski Gasket and open GL control functions.

5.

(a)

(b)

(c)

Write notes on RGB and HSV color models?

What are the various logical graphics input primitives? What is the various input modes in which they work?

TCS-502

1009

srl & palau

Odd Semester Examination 2018-19

w

B.TECH (CSE) (SEMESTER-V)

COMPUTER NETWORK

Time: 03:00 Hours

Max Marks :100

[5×4=20]

What is the computer Network? What is the Network structure and architecture?

What is the OSI reference model? Write about various classes of IP addresses?

1.

Attempt **any four** :

(a)

(b)

(c)

(d)

(e)

What are the advantages and disadvantage of transport layer?

Differentiate between OSI and TCP/IP reference models.

Describe the working principle of Carrier sense multiple access with collision

Detection (CSMA/CD).

2. Attempt **any four**:

(a)

(b)

[5×4=20]

Explain the Local Access Network Design in details. What are the name the four basic network topologies and cite an advantage of each type.

Differentiate between Flow control and Error control. What is the difference between a port address, a logical address and a physical address?

(c)

Discuss different standards of Ethernet.

(d)

(e)

Write difference between static and dynamic routing and Explain dijkstra routing algorithm.

Differentiate between Circuit Switching, Packet Switching and Message switching.

TCS-502/2180

(1)

[P.T.O.]

3.

Attempt **any two**:

[10×2=20]

(a)

A low transmission tower used to transmit data using 3 KHz bandwidth over the link along with 1 Watt of noise power. Calculate the rate of transmission.

(b)

Explain the different type of header supported by IPv6. What is the difference between IPv4 and IPv6.

4.

5.

(c)

Explain the functions of session, presentation and application layer in OSI reference model.

Attempt **any two**:

[10×2=20]

(a) An aloha network user 21.2 kbps channel for sending message packet of 100 bit long size. Calculate the maximum throughput for pure ALOHA network.

(b) Explain TCP/IP architecture with a layer diagram. Explain the function and protocol and services of each layer

(c)

What do you mean by transmission impairment? Explain the causes of transmission impairment?

Attempt **any two**:

(a)

[10×2=20]

What do you understand by Sliding Window protocol and Data Link Protocols?

What is point-to-point protocol? With a neat schematic explain the frame structure of PPP protocol.

(b) What is IP Addressing? How it is classified? How is subnet addressing is performed?

(c)

With a neat diagram explain the working of stop-and-wait protocol. What is the need for sequence numbers? What is the reason for its inefficiency and how is it addressed in Go-Back-N ARQ protocol.

TCS-502/2180

(2)

-X.

Sub Code: CST021 ROLL NO.....

EVEN SEMESTER EXAMINATION, 2024 – 25

.207 | 18/06/2025 12:45:55

2nd Year (IV Sem) B.Tech.: CS&E/AI&ML/IT

COMPUTER NETWORKS

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	ST021_005 103.170.70.207 18/06/2025 12:45:55 Answer any two parts of the following. (10x2= 20) a) (i) What are the design issues of network architecture? Explain. (ii) Discuss the devices of each layer of OSI reference model. b) What is OSI model? Use the network architecture knowledge to compare TCP/IP and OSI model. c) Define transmission media. Explain fibre optic cable in brief. Also explain the advantages of optical fibre over twisted pair and coaxial cable.
Q 2.	103.170.70.207 18/06/2025 12:45:55 Answer any two parts of the following. (10x2= 20) C a) (i) Discuss the Point-to-Point protocol with the help of frame format. (ii) What is piggybacking? Suppose we need a dataword of at least 7 bits. Calculate the values of k and n that satisfy this requirement. b) Explain different types of persistence method. How CSMA/CD protocol is different from CSMA/CA protocol? Explain. c) Explain Selective Repeat ARQ protocols. What are the advantages of Selective Repeat ARQ over Go Back-N ARQ protocol?

Q 3.	CST021_005 Answer any two parts of the following. (10x2= 20) a) (i) How is the BOOTP different from DHCP? (ii) Describe the problem of count to infinity associated with distance vector routing technique. 5 12:45:555 b) What are different routing algorithms? Write the implementation of shortest path routing. c) Explain IPv6 header format in detail. Write advantages of Next-generation IPV6 over IPV4
Q 4.	3.170.70.207 18/06/202Answer any two parts of the following. (10x2= 20) a) (i) What do you understand by Quality of Service? Explain various Quality of Service parameters. (ii) Enumerate how the transport layer ensure that the complete message arrives at the destination and in the proper order. b) Differentiate TCP and UDP in context of the header format. c) What is congestion? Briefly describe the techniques that prevent congestion.
Q 5.	CST021_005 10Answer any two parts of the following. (10x2= 20) a) (i) Explain Asymmetric cryptography. Also write the steps used in RSA algorithm. (ii) What are firewalls? Why they are needed? Explain the different types of firewalls. b) Elaborate about TELNET. Discuss its working procedure in detail. c) Discuss the features of HTTP and also discuss how HTTP works.

TCS 502

Roll No.

Odd Semester Examination, 2019-20 B.Tech (CSE, SEMESTER V) COMPUTER NETWORKS

Time: 3 Hours

Note: (i)

(ii)

Attempt ALL questions.

All Questions carry equal marks. (iii) Be precise in your answer.

Q.1. Attempt *any four* parts of the following:

Total Marks: 100 Total no. of printed pages: 1

(5 × 4 = 20)

(a)

(b)

(c)

(d)

(e)

What are reasons for using layered protocols?

What is ISDN? Draw the ISDN communication architecture? Differentiate between twisted pair, co-axial and fiber optic cable.

If the frequency spectrum of a signal has a bandwidth of 1600 Hz with the highest frequency at 800 Hz according to the Nyquist theorem. What should be the sampling rate?

Write short notes on the following term:

i) MAN ii)

WAN

Q.2. Attempt *any four* parts of the following:

(5 × 4 = 20)

(a)

Compare and contrast CSMA/CD with CSMA/CA.

(b)

What is IEEE standards used for lan and wireless explain?

(c)

(d)

i)

Sketch the Manchester encoding to the bit stream: 0101110101. State drawbacks of stop and wait protocol.

Explain performance issues for the following data link layer protocols:

Go-Back-N protocol.

Q.3.

(a)

(b)

(c)

Q.4.

(a)

ii)

Sliding window protocol

Attempt any two parts of the following:

(10 × 2 = 20)

Explain the congestion control mechanisms to regulate the traffic in a data network.

Discuss RPC and various design issues of presentation layer and transport layer. Explain routing and bellman ford algorithm.

Attempt any two parts of the following:

(10 × 2 = 20)

What is UDP? What is the maximum and minimum size of a UDP datagram? Also

discuss the use of UDP.

(b)

What is cryptography? Differentiate between symmetric key cryptography and asymmetric-key cryptography.

(c)

What is silly window syndrome in TCP flow control? Explain Nagel's algorithm and Clark's solution for the syndrome created by sender and receiver.

Q.5. Attempt *any two* parts of the following:

(a)

(i)

(ii)

(10 × 2 = 20) Define the various data compression techniques used in network? What is the difference between http and https protocol?

(b)

(i)

(ii)

How does FTP work? Differentiate between passive and active FTP. What are the problems for full implementation of voice over IP? Do you think we will stop using the telephone network very soon?

(c)

(i)

(ii)

What is multipurpose Internet Mail Extension (MIME) and for what it is used?

Compare and contrast TCP with RTP. Are both doing the same things?

1

S0029

Sub Code: AHT009 ROLL NO.....

ODD SEMESTER EXAMINATION, 2024 – 25

3rd Year (5th Sem) B.Tech.

CONSTITUTION OF INDIA

Duration: 2:00 hrs Max Marks: 50 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. a) Explain the following terms: i) Rule of Law ii) Federal System of Government. b) Discuss the salient features of the Indian Constitution. c) Explain the significance of Preamble Indian Constitution.	5x2=10
Q 2.	Answer any two parts of the following. a) What do you understand by Directive Principles of State Policy? b) Discuss the Fundamental Rights conferred to citizens as per the Indian Constitution. c) What are the Fundamental Duties as defined in the Indian constitution?	5x2=10
Q 3.	Answer any two parts of the following. a) Explain the Parliamentary form of Indian Government. b) What are the directives of National Emergency and President Rule as per Indian Constitution? c) Discuss the process of Amendment of the Constitutional Powers as defined in the Constitution.	5x2= 10

Q 4.	Answer any two parts of the following. a) In the Framework of the Central Government, discuss the significance of the President. b) Explain the framework of the Local self-Government with its major functionaries. c) How is the framework of the State Government different than the Union Territories?	5x2= 10
Q 5.	Answer any two parts of the following. a) Explain the significance of the Constitutional bodies like UPSC and Election Commission. b) Explain the significance of Non-Constitutional bodies like NITI Aayog and CBI. c) Discuss the role and significance of Cooperative Societies.	5x2= 10

Sub Code: AHT009 ROLL NO.....

ODD SEMESTER EXAMINATION, 2024 – 25

3rd Year (5th Sem) B.Tech.

CONSTITUTION OF INDIA

Duration: 2:00 hrs Max Marks: 50 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. a) Explain the following terms: i) Rule of Law ii) Federal System of Government. b) Discuss the salient features of the Indian Constitution. c) Explain the significance of Preamble Indian Constitution.	5x2=10
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Q 3.	Answer any two parts of the following. a) Explain the Parliamentary form of Indian Government. b) What are the directives of National Emergency and President Rule as per Indian Constitution? c) Discuss the process of Amendment of the Constitutional Powers as defined in the Constitution.	5x2= 10
Q 4.	Answer any two parts of the following. a) In the Framework of the Central Government, discuss the significance of the President. b) Explain the framework of the Local self-Government with its major functionaries.	5x2= 10

	c) How is the framework of the State Government different than the Union Territories?	
Q 5.	Answer any two parts of the following. a) Explain the significance of the Constitutional bodies like UPSC and Election Commission. b) Explain the significance of Non-Constitutional bodies like NITI Aayog and CBI. c) Discuss the role and significance of Cooperative Societies.	5x2= 10

* 1 0 7 3 *

PAPER ID : 1073

TCS-501

Printed **Pages:** 3

Paper ID and Roll No. **to be filled in your Answer Book** Roll No.

B. Tech.

(SEM. V) (ODD SEM.) (REG. & BACK) EXAMINATION, 2012-13

COMPUTER GRAPHIC

Time: **3 Hours]**

[Total Marks: **100**

UNIT - I

1

Do any **four:**

(a)

10731

(b)

5×4=20

List the operating characteristics for the following display technologies: raster refresh system, vector refresh system, plasma panel and LCD. How much time is spent scanning across each row of pixels during screen refresh on a raster system with a resolution of 1280 by 1024 and a refresh rate of 60 frames per second?

(c) Implement the polynomial function using the DDA algorithm, given any number (n) of input points is to be plotted when n=1.

(d) Implement the line type function by modifying Bresenham's line drawing algorithm to display either solid, dashed, or dotted lines.

(e)

(f)

List some applications appropriate for each of the display technologies.

List the different input and output components that are typically used with virtual reality systems. Also explain how users interact with a virtual scene displayed with different output devices.

1

[Contd...

UNIT - II

UNIT - IV

2

Do any two:

(a)

(c)

4

Do any **two**

10×2=20

Set up a geometric data table for unit cube using only: (a) vertex and polygon tables, and (b) a single polygon table. Compare the two methods for representing the unit cube with a representation using three data tables, and estimate storage requirements for each.

oor (b) Define an efficient polygon representation for a cylinder. Justify your choice of representation. How would the values for parameters A,B, C, and D in the equation of plane surface have to be altered if the coordinate reference is changed from a right handed system to left handed system? Explain in detail.

UNIT - III

(a)

10×2=20

Derive the transformation matrix for scaling an object by a scaling factor s in direction defined by direction angles a , b and c .

(b) Using any clipping procedure, write a procedure to perform a complete viewing transformation from world coordinates to device coordinates for any specified parallel projection vector.

(c) Explain a back face detection procedure using a perspective projection to view visible faces of a convex polyhedron. Assume that all parts of object are in front of the view plane, and provide a mapping onto a screen view-port for display.

UNIT - V

3

Do any two:

(a)

1073]

5

Do any **two**:

(a)

10×2=20

Prove that the multiplication of transformation matrices for each of the following sequence of operations is commutative:

(i)

Two successive rotation

(ii) Two successive translation

(b) What are the results of performing two successive block transfers into the same area of a frame buffer using the binary arithmetic operations? Write a routine to implement rotation by any specified angle in a frame buffer block transfers. Compare the number of arithmetic operations performed in the Cohen-Sutherland and the Liang- Barsky line clipping algorithms for several different line orientations relative to clipping window.

(c)

2

(b)

10×2=20

Write a program to implement dynamic motion specifications. Specify a scene with two or more objects, initial motion parameters, and specified forces.

Write an interactive procedure that allows selection of HSV color parameters from a displayed menu, then the HSV values are to be converted to RGB values for storage in a frame buffer.

(c) Write short notes on following:

(i) (ii)

OpenGL API

The Sierpinski Gasket.

1073]

3

[2500]

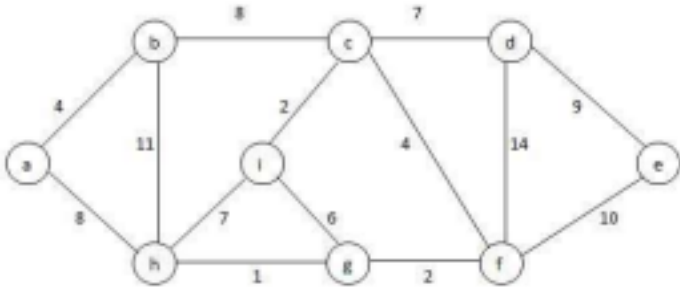
[Contd...

Sub Code: BCST 503 ROLL NO.....

Vth SEMESTER EXAMINATION, 2023 – 24

IIIrd Year, B.Tech. – Computer Science & Engg/Information Technology DESIGN AND ANALYSIS OF ALGORITHMS

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) Differentiate between backtracking & branch and bound technique. b) Show that vertex cover problem is NP-Complete. c) Use Prim's algorithm to find the minimum cost of spanning tree of the following graph.  d) Give the solution to the m-coloring of a graph using backtracking. e) Explain Merge sort technique. Discuss the time complexity of merge sort. f) Explain Strassen's matrix multiplication and its time complexity.	5x4=20
Q 2.	Answer any four parts of the following. a) Write and explain the Cook's theorem. b) How can the output of the Floyd Warshall algorithm be used to detect the presence of a negative weight cycle? c) Compare and contrast between connected components and bi connected components. d) How 8-Queen's problem can be solved using back tracking? Explain. e) Solve the recurrence relation and find the time complexity of $T(n)=7T(n/2)+18n^2$. f) Explain AND/OR graphs.	5x4=20

Q 3.	Answer any two parts of the following. a) Discuss in detail about the class P, NP, NP-hard and NP-complete problems. Give examples for each class. b) Define Greedy knapsack problem. Find the optimal solution of the Knapsack instance $n=7$, $M=15$, $(p_1, p_2, \dots, p_7) = (10, 5, 15, 7, 6, 18, 3)$ and $(w_1, w_2, \dots, w_7) = (2, 3, 5, 7, 1, 4, 1)$. c) Define Job sequencing with deadlines. What is the solution generated by the function JS when $n = 7$, $(p_1, p_2, \dots, p_7) = (3, 5, 20, 18, 1, 6, 30)$, and $(d_1, d_2, \dots, d_7) = (1, 3, 4, 3, 2, 1, 2)$?	10x2= 20
Q 4.	Answer any two parts of the following. a) What is dynamic programming? Find the order of parenthesization for the optimal chain multiplication of the following matrices: $A_1 = 15 \times 5$ $A_2 = 5 \times 10$ $A_3 = 10 \times 20$ $A_4 = 20 \times 25$	10x2= 20

	b) What is difference between Prim's and Kruskal's algorithm? How do you construct a minimum Spanning tree using Kruskal's algorithm? Explain. List any two applications. c) Using the procedure of quick-sort algorithm to sort the following elements: 2,8,7,1,3,5,6,4. Also define the complexity of quick-sort.	
Q 5.	Answer any two parts of the following. a) Apply branch and bound method to solve travelling salesman problem for the graph whose cost matrix given below: ∞ 7 3 12 8 3 ∞ 6 14 9 Cost matrix = 5 8 ∞ 6 18 9 3 5 ∞ 11 18 14 9 8 ∞ b) Explain how solution will be provided for all pairs shortest path problem using dynamic programming.	10x2= 20

	c) What is randomized algorithm? Write advantage and disadvantage of randomized algorithm. Also explain the types of randomized algorithm.	
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Sub Code: BCST 502 ROLL NO.....

Vth SEMESTER EXAMINATION, 2023 – 24

IIIrd Year, B.Tech. – Computer Science and Engineering/Information Technology
COMPUTER NETWORKS

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) What are two reasons for using layered protocols? What is one possible disadvantage of using layered protocols? b) Suppose a computer sends a packet at the network layer to another computer somewhere in the Internet. The logical destination address of the packet is corrupted. What happens to the packet? How can the source computer be informed of the situation? Explain. c) State the requirements of CRC. d) What is IP addressing? How it is classified? How is subnet addressing is performed? e) Explain the SNMP protocols in detail. f) Explain the concept of File Transfer Protocol (FTP) with the help of diagram in application Layer.	5x4=20
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Q 2.	Answer any four parts of the following. a) Describe HDLC with the help of frame format. b) The power of a signal is 10mW and the power of the noise is 1μW. What are the values of SNR and SNRdB. c) Differentiate between ARP and RARP protocol. d) Describe the problem of count to infinity associated with distance vector routing technique. e) What is congestion? Describe the techniques that prevent congestion. f) Enumerate how the transport layer ensure that the complete message arrives at the destination and in the proper order.	5x4=20
Q 3.	Answer any two parts of the following. a) Compare and contrast TCP/IP with OSI model. b) Briefly describe the CSMA/CD. Explain the similarities and differences between CSMA/CD and ALOHA protocol. c) Explain about the TCP header and working of TCP protocol and differentiate between TCP and UDP with frame format.	10x2= 20
Q 4.	Answer any two parts of the following. a) What is transmission media? How guided media differ from unguided media? Explain co-axial cable in brief. b) Explain IPv4 with the help of header format. Also explain the reason of transition from IPv4 to IPv6. c) If the 7-bit Hamming codeword received by a receiver is 1110101. Assuming the even parity, state whether the received codeword is correct or incorrect. If incorrect, locate the position of error. What will be the correct code?	10x2= 20
Q 5.	Answer any two parts of the following.	10x2= 20

- | |
|---|
| <p>a) Encode the data stream 10011010 using the following encoding scheme: i)
Bipolar NRZ-L</p> <p>ii) Bipolar NRZ-I</p> <p>iii) RZ</p> <p>iv) Manchester</p> <p>v) Differential Manchester</p> <p>b) Explain Sliding window data link control protocols. How selective repeat ARQ is different from GO-BACK-N.</p> <p>c) What is unicast routing? Discuss unicast routing protocols.</p> |
|---|

TCS-603

152

Roll No. to be filled in your Answer Book

Soipol-Roll No.

Printed Pages: 4

along with B. Tech. (CS) (6TH Semester) (b)

Examination, 2015

Artificial Intelligence

Time: 3.00 Hrs]

Note: Attempt all questions.

[Max. Marks: 100

Q1. Attempt any four questions from the following: (5x4 = 20)

mel

bro(a) Illustrate the steepest-ascent hill climbing technique in detail.

(b) List down the characteristics of intelligent agent.

anovieve

Batec(c) Explain A* algorithm with an example. State the limitation in the algorithm.

(d) Explain the concept of planning with state space search using suitable examples.

(e) What do you mean by local maxima with respect to search technique?

(1)

Distinguish between state space search and plan **space** search.

Q2. Attempt any four questions from the following: (5x4 = 20)

(a) Explain in detail learning from observation and explanation based learning.

(1)

TCS-603/2000/4

(6)

и

(b) What is the use of online search agents in unknown

(c)

environment? Hood

buoy ni beli

How TELL and ASK are used in first-order-logic?

(d) Give the PEAS description of an interactive English tutor system.

(e) Define artificial intelligence.

(1) How can you represent the resolution of predicate oorgeeMlogic?

TenH 00.8

Q3. Attempt any two questions from the following:

эти

(10×2=20)

(a) Analyse the missionaries and Cannibals problem

which is stated as follows. 3 missionaries and 3

cannibals are on one side of the river along with a boat that can hold one or two people.

Find a way to get everyone to the other side, without leaving a group of missionaries in one place outnumbered by the cannibals in that place.

(b) Explain the constraint satisfaction producer to solve the crypt arithmetic problem.

CROSS

+ ROADS

DANGER

nclo brie dospeles

upindost

newed en

(c) Explain the functional local search strategies with example:

(i)

genetic algorithms

DAS

1.000

(ii) desimulated annealing

(iii) local beam search

(2)

Q4. Attempt any two questions from the following:

(a) Write short notes on:tor

(i) Temporal models

messigst

(10×2 = 20)

pninuig sted.srialAbos partinoplAxMM nisiqxd (d)

(ii) Probabilistic Reasoning

23 ne la atrenogmoo

Explain the following uniformed search strategies:

(b)

23 edi ni be

(1)

Depth First Search

(ii)

Iterative deepening depth-first search

(iii) Bidirectional search

(c) Consider the following sentences:

john likes all kinds of food

apples are food

chicken is food

anything anyone isn't killed by is food.

Bill eats peanuts and is still alive

Sue eats everything bill eats

(i)

Translate these sentences into formulas in predicate logic

(ii) Prove that john likes peanuts using backward chaining

(iii) Convert the formulas of a part into clause form

(iv) Prove that john likes peanuts using resolution

(3)

P.T.O.

Q5. Attempt any two questions from the following:

($10 \times 2 = 20$)

(a) Explain the concept of Bayesian network in representing knowledge in an uncertain domain.

(b) Explain Min-Max Algorithm and Alpha-beta pruning.

(c) List and define the major components of an ES. Describe how expert systems perform inference. What are the major activities performed in the ES blackboard (workplace)?

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TCS-603/2000/4

TCS 604

96

Printed Pages: 2

Paper ID and Roll No. to be filled in your Answer Book

Roll No.

B.Tech (SEM.VI)

Examination, 2015

GRAPH THEORY

Time: **2 Hours**

Total Marks: **50**

NOTE: ALL QUESTIONS ARE COMPULSORY. Q.1. Attempt any four of the following:

22 x 4

- (a) Prove that a simple graph with n vertices and k components can have at most $[(n-k)(n-k+1)]/2$ edges.
- (b) Define Hamiltonian Circuit and Hamiltonian path with suitable example. (c) Define matching. How it is related to bipartite graph?
- (d) Show that every tree with two or more vertices is 2-chromatic.
- (e) Differentiate between incidence and adjacency matrix.

Q.2. Attempt any four of the following:

- (a) What is a Graph? Draw a complete graph with $n=4$.
- (b) Draw all simple graphs of five vertices.

22 x 4

- (c) Prove that there is one and only one path between every pair of vertices in a tree T .
- (d) Perform ring-sum on two graphs and draw the final graph obtained.
- (e) Calculate degree for all the vertices in the given graph:

V1

V4

V2

(1)

TCS-604 // 2000

Q.3. Attempt any two of the following:

5+5

- (a) What is isomorphism? How is it connected to 1-isomorphism and 2-isomorphism?
- (b) Explain Elementary reduction method used to prove a graph to be planar or non-planar.
- (c) Discuss the concept of Matching and Chromatic number used in coloring a graph. Elaborate your answer with suitable examples.

Q.4. Attempt any two of the following:

5+5

- (a) What is Separable Graph? What is the importance of Connectivity in separable graph?
- (b) What are the steps followed in Prim's algorithm to find minimal Spanning tree? Find the same in the graph given below:

[4 x 5=20]

V1

V4

7

V2

5

5

V3

[4 x 5=20]

- (c) Prove the Euler formula for number of regions for all the connected

planar graphs.

Q.5. Attempt any two of the following:

5+5

(a) What is the importance of Travelling Salesman Problem in Graph Theory?

(b) What do you understand by incidence matrix? Take a graph and draw its incidence matrix.

(c) Draw Circuit Matrix and Fundamental Circuit Matrix in the given graph.

E1

V1

E4

V4

14

55

V2

E6

E2

V3

E3

(2)

TCS-604 // 2000

TCS-601

9

Roll No. to be filled in your Answer Book

Roll No.

Printed Pages: 4

B. Tech. (CS) (6TH Semester)

no

Examination, 2015

Operating System

Time: 3.00 Hrs]

Note: All questions are compulsory.

[Max. Marks: 100] failing with

balld (b)

Q1. Attempt any four parts:

(5×4 = 20)

(a) What is operating system? Mention the objectives and functions of operating system.

(as (b) Explain process control block.

107

80

(c) Describe the long term, short term and medium term schedulers.

What is Multitasking and real time Operating System?

(e) Explain the term Kernel and Shell.

(1) Explain the security and protection provisions implemented in UNIX operating system.

(1)

TCS-601/1970/4

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Q2. Attempt any four parts:

(a)

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(5×4 = 20)

Describe the preemptive and non-preemptive scheduling. Explain each type with an example.

(b) Explain segmentation in detail. Compare it with paging.

(c) What is Context switching?

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(d) Discuss in detail the concept of virtual machines, @qmoo eis anorteaup Aston with neat sketch.

(OS (e) Discuss threading issues which are considered with multithreaded programs.

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(1) Explain solutions of recovery in detail.

Q3. Attempt any two parts:

(10×2=20)

Amex (a) Suppose the following processes arrive for execution at the time indicated:

enitens q

Process

Arrival time

Burst time

P1

0

8

P2

1

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enjalvord noil P3

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Draw the Gantt chart and calculate the average time and turn around time for the processes with SJF preemptive and SJF non-preemptive scheduling algorithm.

(b) What are the approaches that can be used for prevention of deadlock?

(c) Explain FCFS and Round Robin scheduling with suitable example.

Q4. Attempt any two parts:

98

(10x2 = 20)

(a) What is seek time and rotational latency? Explain shortest seek time first Disk scheduling.

(b) What is thrashing? When does it occur? Explain.

(c) Why should page replacement be performed?

Explain FIFO policy in detail.

Q5. Attempt any two parts:

al

($10 \times 2 = 20$)

(a) Give a solution to dining philosopher's problem and explain.

(b) Given memory partitions of 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB (in order), how would each of the first-fit, best-fit, and worst-fit algorithms place processes of 212 KB, 417 KB, 112 KB, and 426 KB (in order)? Which algorithm makes the most efficient use of memory?

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CS-TCS-601/1970/4

(c) Consider the following snapshot of a system:

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Allocation.

Max

Available

A B C D A B C D

A B C D

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P1

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ritw.pril P2 1354

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P3 0632

0 6 5 2

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Answer the following questions using banker's

algorithm:

Sbermo (i) What is the content of the matrix Need?

(ii) Is the system in a safe state?

(OS=S-ON)

(iii) If a request from process P1 arrives for bris meldong e10

(0, 4, 2, 0) can the request be granted

immediately?

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Roll No.

B. Tech. 4th Semester CSE

UTU (EVEN SEM.-IV) Examination-2015

Write a shell

UNIX AND SHELL PROGRAMMING

Time: 2 Hrs.]

Note: Attempt all questions.

1.

Attempt any three parts.

ame then

[Max. Marks:50

Sighoa lleda

1212) $5 \times 3 = 15$

a) List and explain various modes of vi editor. Give some commands to operate with these modes.

b) How many types of files are there in Unix explain each of them.

c)

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What is the role of kernel and shell in Unix?

d) Explain the following commands with their options.

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Attempt

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Attempt any three parts.

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a) Explain the difference between following commands?

UNIX He

(1) cmp and diff \$2 rbe.dbst.8

оя

in

(2) dos2unix and unix2dos) 'UTU

b) Explain the positional parameters in Unix? How S

02:20 you ac

you access command line arguments from within a shell script?

c) Explain the following commands?

ar=exa (1) \$ls-1

Tanoitesup listqmqstA:stoM

eheq sendt vris igmatA

evi noti (ii) v \$ zip -d backupt.zip myfile.dat tei. (e

esbor (iii) \$ cat-n myfile.txt abismo mo

nisiqxex (iv) \$ (date; who) >> myfile.txt sim woh: td

(v) \$ who-q

ment to rose

Sxinu ni llera bos lemed to eli si ai terW (o d) Explain the facility of mounting and un-mounting t

eniwollot erit nisiqx3 (b

anoilgofiles in Unix.

3.

Attempt any two parts:

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5x2=10

bo, (S)

a) Write a shell script to create the following menu and perform the operation as per the given choices using case statement.

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TCSbold'sisb br Main Menu

Paper Code

(1) all Copy File

Roll No (2) Remove File b

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(3) Remove Directory matava elit XIMU

UTU (4) Exi

(4) Exit From Program

b) Write a shell scrip which receives two file names UNTY AND

as arguments. It should check whether the two file

Time: 21 contents are same or not. If they are same then

Note:

remove both files.

estions.

c) Explain the following commands with their options:

by three

(1) cat

some.com.

(2) zip and unzip

Gar (3) mailx

editor. Give

hese

s-in Unix explain

(4)

cp

shell in Unix?

4.

Attempt any two parts:

$$5 \times 2 = 10$$

a) Explain the commonly used Shell Metacharacters with an examples.

(3)

P.T.O.

b) What is a Boot block, super block and data block?

Explain Unix boot process. yooo

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c) Explain backup and recovery related commands in UNIX file system. not

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Roll No. to be filled in your Answer Book

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B.Tech. (CS) (6TH Semester)

Examination, 2015lxlad

Visual Programming and .Net Technologies 10 consistlib bris 29inslimia ent
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Time: 3.00 Hrs]

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Q1.6 Attempt any four parts:

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[Max. Marks: 100

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egy consist

(a) How is the Common Language Runtime different
from other runtimes available?

Seboe

(b) What are different types of JIT?

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(b)

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(c) Explain the components of .NET Framework. Draw ties with
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a neat diagram for the same.

s ei teriW

(d) What is an Assembly? What are the different types

S

of assembly? If you want to view an assembly how delay
would you go about it?

(e) Define the following:

aldetive (i) fi Namespace in C#.

Lis

(ii) Assembly manifest

demettAED

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TCS-605/000/3 1980

(f) b Explain about protected internal access specifier.

Q2. Attempt any four parts:

(5×4 = 20)

(a) What is the concept of boxing and unboxing?

Explain it with suitable examples.

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(b) Briefly summarize the similarities and differences
between arrays and collections.

(OS= (c)) List the differences between a value type and a
reference type.

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(d) Is it possible to have two Main() in a C# code? If

so, how is it resolved?

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(e) What is the difference between String and String

Builder classes? Explain.

(f) What is an Indexer in C#? Implement a sample

tib edi worl vidms indexer.

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Q3. Attempt any two parts:

(10×2=20)

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(a) What is the .Net runtime's way of dealing with

unhandled exceptions? Explain it with suitable

example.

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(b) What do you mean by Encapsulation? Show the use

of properties for it. Discuss the role of accessors in

defining properties of a class.

(c) Explain about protected internal access specifier.

Q4. Attempt any two parts:

(a) Define the following:

(i)

Adhoc Distruction method

(10x2 = 20)

(ii) System.Gc

How (b) Explain the following in context with interface:

Multicast delegates

(ii) Cloneable object

(c) What is Interface? Explain interface hierarchies with suitable example.

Q5. Attempt any two parts:

mework Draw

(10x2 = 20)

(a) What are Shared Assemblies? Also explain delay signing installing and removing shared assembly.

(b) What is DLL Hell problem? Explain.

(c) List and explain the members of System.Object class.

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TCS-3/1980/5

BCST-602

Roll No.

Even Semester Examination, 2021-22

Course Name: B.TECH

Branch: CS

Semester: VI

Subject: Compiler Design

Time: 3 Hours

Max Marks: 100

Number of Printed pages: 2

Note: - Attempt all questions: All Questions carry equal marks

Q1. Attempt any four parts of the following

- (a) Define compiler with some construction tools
- (b) Differentiate between compiler and interpreter.
- (c) Summarize function of preprocessor?

(5 x 4 = 20)

- (d) What is lexical analysis discuss the design of a lexical analyzer generator?
- (e) Explain LEX compiler

Q 2. Attempt any four parts of the following

- (a) Define Context Free Grammar with example
- (b) Define parser, with its advantages and disadvantages

(5 x 4 = 20)

- (c) What is meant by syntax tree? Explain the construction of syntax tree with an example?
- (d) Explain Top Down parsing and compare it with Bottom Up parsing
- (e) What are S-attributed and L-attributed definition?

Q3. Attempt any two parts of the following

- (a) Describe the Error recovery scheme in YACC

(10 x 2 = 20)

(b) What is Code Optimization? Mention the criteria for achieving machine independent optimization.

(c) Design LL(1) parsing table for following:

R-251

A

→ AcB/CC/C

B

→ bB/id

C

→ CaB/BbB/B

1

P.T.O.

Q4.

Attempt any two parts of the following:

(10 x 2 = 20)

(a) Draw NFA for the regular expression $ab^* \mid ab$

(b) Construct the minimised DFA for the regular expression $(0+1)^*(0+1) 10$.

(c) Construct and explain the DAG for the following basic block.

d := b * c

e := a + b

b := b * c

a := e - d.

Q5. Attempt any two parts of the following:

(a) Explain formal grammar with suitable example

(b) Write short note on LALR and SLR parsing tables

(c) Explain symbol table with proper examples

R-251

2

(10 x 2 = 20)

BCST-603

Roll No.

Even Semester Examination, 2021-22

Course Name : B.TECH

Branch: CSE/IT

Semester: VI

Subject: Data Analytics

Time: 3 Hours

Note: - Attempt all questions: All Questions carry equal marks

Max Marks: 100

Number of Printed pages: 1

Q 1. Attempt any four parts of the following

(5 x 4 = 20)

(a) Describe the structure of HDFS in a Hadoop ecosystem using a diagram. (b) Describe the bootstrapping and its importance

(c) What is sampling and sampling distribution giving a detailed analysis.

(d) What is the significance of scatter plot matrix?

(e) Why do we need Bloom filter in filtering streams?

Q 2. Attempt any four parts of the following

a) What are the strength & weakness of CLIQUE

(5 x 4 = 20)

b) Explain Grouping, Join, Co Group, Cross & Group in data

c) List out and explain the data types in Hive.

d) Discuss in brief about Mapper code and Reducer code e) Explain the process Crowd Sourcing Analytics

Q 3.

Q4.

Attempt any two parts of the following

(10 x 2 = 20)

(a) Explain Google File System architecture with neat diagram (b) Discuss Pig Latin Application Flow.

(c) Describe any two-sampling technique for Big Data with the help of Examples

Attempt any two parts of the following:

(10 x 2 = 20)

(a) What is Cluster? Explain the setting up a Hadoop cluster

(b) Illustrate YARN based execution model and its functions With a neat diagram (c) Discuss Analysis of Variances (ANOVA) and correlation indicators of linear relationship

Q5. Attempt any two parts of the following:

(10 x 2 = 20)

(a) Write in detail the concept of developing the Map Reduce Application

(b) What are the various stages in big data analytics life cycle? Illustrate with a figure, explaining each of them

(c) With an example, explain the term social media analytics.

R-252

BCST-604A

Roll No.

Even Semester Examination, 2021-22 **Course Name : B.TECH**

Branch: Computer Science & Engineering

Semester: VI

Subject: Graph Theory

Time: 3 Hours

Max Marks: 100

Number of Printed pages: 3

Note: - Attempt all questions: All Questions carry equal marks

Q1. Attempt any four parts of the following

(a) Define **the** following with one example each:

i) Subgraph

ii) Spanning **Subgraph**

(b) Prove that in a graph **the** number of the vertices with odd degree is **even**.

(c) Define **the** Hamiltonian path with example.

(5 x 4 = 20)

(d) Define isomorphism between two graphs. Are the following graphs are isomorphic to each other? Justify your

answer.

(e) Discuss **the** dual of a subgraph with example.

Q 2. Attempt any four parts of the following

(a) Draw the adjacency graph for the following adjacency matrix

01010 10011

$X(G)=00011$

11101 01110

(5 x 4 = 20)

(b) Prove that a connected planar graph with n vertices and e edges has $e-n+2$ regions.

(c) Prove that in a tree $T(V,E)$, $|V|=|E|+1$.

(d) Show that in any group of two or more people, there are always two with exactly the same

number of friends inside the group.

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(e) For a Eulerian graph G , prove the following properties.

i)

The degree of each vertex of G is even. ii) G is an edge-disjoint union of cycles.

Q3. Attempt any two parts of the following

(10 x 2 = 20)

(a) Consider the following graph G and any one of its spanning trees, T . List all fundamental

circuits and fundamental cut-sets with respect to T .

e5

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b

(b) Discuss an algorithm for finding the connected components of a graph G with suitable example. (c) Explain Chromatic Partitioning in detail.

Q4.

Attempt any two parts of the following:

(10 x 2 = 20)

(a) Discuss an algorithm for finding the shortest path from a specified vertex to another specified

vertex. Illustrate with example.

(b) Explain directed paths and Connectedness.

(c) Write Dijkstra's Shortest path algorithm and apply this algorithm to find the shortest path.

Q5. Attempt any two parts of the following:

(10 x 2 = 20)

(a) Give the incidence matrix of the graph G. Also write the properties of incidence matrix.

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(b) Define Spanning tree. Find any two spanning trees T1, T2 of the graph G given below. Also

find the branch set, chord set, rank and nullity.

a

e

(c) Draw a flowchart indicating all the five conditions to find the spanning tree /spanning forest.

Apply it to find the spanning tree /spanning forest for any disconnected graph of your choice.

BCST-601

Roll No.

Even Semester Examination, 2021-22

Course Name : B.TECH

Branch: CSE/IT

Time: 3 Hours

Semester: VI

Subject: Microprocessors and Applications

Max Marks: 100

Number of Printed pages: 1

Note: - Attempt all questions: All Questions carry equal marks

Q1. Attempt any four parts of the following

(a) Explain Minimum mode of **8086** MP with neat diagram.

(5 x 4 = 20)

(b) Explain in detail with a neat figure the working of the internal architecture of the **8086** MP.

(c) Define the RISC and **CISC** processor.

(d) Explain XCHG and CMP instructions.

(e) Define addressing mode of **8086**.

Q 2. Attempt any four parts of the following

(5 x 4 = 20)

(a) With neat diagram explain the function of each component: microprocessor, memory, I/O and Buses. (b) Enlist the advantages of assembly language programming over machine language.

(c) Explain the clock oscillator chip **8284**.

(d) Draw and explain the Intel **8282** Octal Latch chip.

(e) Explain the INC & DEC instructions.

Q3. Attempt any two parts of the following

(a) Write short note on **80286** and Pentium processors.

(10 X 2 =20)

(b) Explain Maximum mode of 8086 MP with neat diagram. Also explain the write timing cycle of

maximum mode.

(c) Explain Direct Addressing, Immediate addressing and Register Relative addressing mode in

reference to 8086.

Q4.

Attempt any two parts of the following:

(10 x 2 = 20)

(a) Draw block diagram of 8253/54 and write how it is initialized by microprocessor.

(b) Draw the pin configuration of 8051. Explain the function of each pin.

(c) Explain the use of microcontrollers in embedded system.

Q5. Attempt any two parts of the following:

(10 x 2 = 20)

(a) Interface 128 KB memory with 8086 using proper diagram and addressing scheme. The Size of each chip of memory is 16 KB. Show the neat sketch with all address lines and necessary hardware with starting address of 40000 H.

(b) Describe the interfacing of an **8259** PIC to **8086** microprocessor.

(c) Explain the additional feature of **80186** microprocessor. What are the differences between

80386 and **80486** microprocessor?

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PAPER ID: 1079

TCS-603

Printed Pages: 3

Paper ID and Roll No. to be filled in your Answer Book

Roll No.

B. Tech.

(SEM. VI) EXAMINATION, 2012

ARTIFICIAL INTELLIGENCE

Time: **3 Hours]**

Note :

[Total Marks: **100**

Attempt all questions. All questions carry equal His marks.

1

Attempt any four parts:

(ii)

2

5×4=20

(a) Explain artificial intelligence along with its historical developments.

(b)

What is a Turing machine? Explain how one can check the intelligence of a machine?

(a)

(c)

Explain various methods for problem representation in artificial intelligence.

Labie odio si of

(d) bs What is an intelligent agent? Explain its architecture with suitable diagram. In qu

(e)

Discuss the impact of AI on society? Explain your answer with suitable reasoning.qmoti

5×4=20

Attempt any four parts: to anoitstimil bos (a) Explain the role of context free and transformational grammars in natural language processing with suitable examples.

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(b) Explain transition nets and augmented nets in understanding of natural languages with suitable examples.

(c)

Explain various steps in natural language processing with suitable diagram and examples.

(d) Explain transducers and parsers to manipulate linguistic knowledge with suitable examples. Explain how fifth generation computer systems are able to natural language processing.

(e)

Attempt any two parts: `

(c)

Explain various knowledge acquisition strategies with suitable examples.

5

Answer any two parts:

(a)

10×2=20

Explain LISP and PROLOG conditional statements constructs with suitable programming illustration.

(b) Write a backward chaining procedure using PROLOG syntax. What is the role of backward chaining in inference engine of expert system?

Write short notes on four :

10×2=20

910

(c)

(i)

Minimax search procedure

Brionesup A anodeuple ignat1A (a) Explain logic along with its types. Explain all rules of inference with suitable examples.

(b) Convert $(A \rightarrow ((B \& C) \rightarrow D))$ into DNF

no

(disjunctive normal form) and $((A \rightarrow B) \rightarrow C)$ into CNF (conjunctive normal form)

(c) Three missionaries and three cannibals are on πou

one side of a river, along with a boat that can hold at most 2. Find a way to get everyone to **the** other side, without ever leaving a group So of missionaries in one place outnumbered by the cannibals in that place.

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Attempt any two parts:

10×2=20

newens Oy

(a) What is expert system? Explain advantages and limitations of these days expert systems.

(b) Name various widely used expert system tools. diw Explain any two of them in detail.

(ii) Knowledge acquisition

(iii) Ant colony system

(iv) Neural Net learning

(v) Bayesian Networks.

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5

Attempt any two of the following questions:

10×2=20

(a) What is global data flow analysis? What is its use

s

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in code optimization? (b) Discuss the loop optimization technique with the help of suitable examples. (c) Consider the following basic block. Draw the DAG representation of the block and identify local common sub expressions. Eliminate the common expressions and rewrite the basic block. Assume that none of the temporary variables (S1, S2, S9) are needed outside the block. The basic block statements are:

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LIO: $SI = 4 * I$

$S2 = \text{addr}(A)$

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$S3 = S2[S1]$ $S4 = 4 * 1$

$S5 = \text{addr}(B) -$

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$S5[S4]$

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$S7 = S3 * S6$

$S8 = \text{PROD} + S7$

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PAPER ID: 1078-N

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Host odt azupeib oaLA SVI0251 915 atsitro DOAY (SEM. VI) EXAMINATION, 2012
gniwoollot to **COMPILER DESIGN (BACK-PAPER)**

Time: **3 Hours]**

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(d)

d [Total Marks: **100** ba

Note: Voll(i) srl Attempt all questions.qqs or sel

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I = S9

If I <= 20 GOTO LIO

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(ii)

All Questions carry equal marks. 101 oldet griz169 12 tobiano

olds

5×4=20

Attempt any four of the following questions: (a) Discuss the phases of compiler with appropriate diagram.

(b) Explain the term cross compiler in detail. OS (c) Construct NFA to DFA for the regular expression 3297bbA-997 to enoitrinemelqmi er 2

119/T_ (a | b)* ab* a. First show the construction of

NFA and then demonstrate DFA construction.

(d + s dotiwa

(d) Differentiate between Macro-processor and Pre-

firgad $y = x \cdot 9280$

\$

processor. How YACC can be used to generate a parser?

(e)

1019 928dq Inoixs. I y bastola poy ob terW (d) abortom taaggua oalA frons 92sily pirose
bris

(f)

2109 to vis

auoiry orlt zaupai

faldet lodmye ei tarW (5)

Also explain the format of input specification file. Explain the action taken by every
phase of the compiler of the following string lustab

oldst lodrive to Rotaxisgio tot bazu zarosong

=

A B C 4 + D/E.

evoda srit not aboo orbbe set art wobethW

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DS-1078-N]

[-8 [1920]

DS-1078-N]

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[Contd...

2 sep Attempt any two of the following questions: T

(a)

(b)

$10 \times 2 = 20$

0(b)

Construct LALR(1) parser table for the grammar $S \rightarrow aA \mid bB \mid cC$

$C \rightarrow \epsilon$

$C \rightarrow b$

OM HOR

Are there any conflict entries? If so how these conflicts are resolved? Also discuss the techniques used to resolve the conflicts in **YACC**.

Design SLR parsing table for following grammar $S \rightarrow aABe$

$\rightarrow$

$A \rightarrow Abc$

00: ahsM A

3

(c)

b

Bd

DOAG)

Write a translation scheme to generate intermediate code for assignment statements with array references. Explain it with the help of suitable parse tree.
notexitq pool or ausai (d) What is basic block? How can a sequence of three-address statements be transformed into a list of blocks? Show with examples. timila enogastqxs, dua nommos lsool

(c)

into ad

Attempt any two of the following questions: (a) Compare static, stack and heap allocations. Goold dfbier in beheer

10×2=20

For the following C program, draw the details of the activation records if

Stack allocation is used

(ii) Heap allocation is used

(show the activation record after 'main' calls 'fun'

returns back to 'main')

edi

4

[auoH & smit

Use the appropriate graph to explain the following to procedure. ups vs aroitasu IIA

Consider SLR Parsing table for:

$S \rightarrow ABlgDa$

$A \rightarrow ablc$

steingoBDc

->>>

$C \rightarrow gClg$

Dg.

Do

Attempt any two of the following questions:

(a) Discuss the implementations of Three-Address to **no** suitable example. Then ts with the help o

noi

(i)

int main()

f

te nisi (d)

int *p;

$10 \times 2 = 20$

p

fun();

}

int *fun()

{

int i

23;

return i;

}

nois consider the following program og ATM

91 he switch(a + b) 6M newid stainsshi -M (b)

begin

case 2: x = y

free case 5: switch xau ed no DAY WOH (6)

begin

slft nometnega un termot et aigips BLA case 0 a b+1

er to case 1: a = b+3st noirs sit nisiqx8 (1) default: x = y+1 wollool **edito** noligmoo end

ENG + \$

Write down **the** Three address code for the above program.

12 01

TAYbbs

\$2

(@hbbs

142122

32

82* 22 = 12

18+ 0099 = 82 доля

82 СОЯЧ

1 1=22 92=1

OLI OTOD OS => [

(b) What do you understand by Lexical phase error and syntactic phase error? Also suggest methods for recovery of errors.

(c)

What is symbol table? Discuss the various approaches used for organization of symbol table.

DS-1078-N]

2

D-[Contd...

DS-1078-N]

3

[Contd...

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5.

a.

TCS-502

Printed Pages: 4

Paper ID and Roll No. to be filled in your Answer Book Roll No.

B. Tech.

(SEM. V) (ODD SEM.) (REG. & BACK) EXAMINATION, 2012-13 **COMPUTER NETWORKS**

b.

c.

UNIT-V

(Do any Two)

10×2=20

Explain how the transfer protocol clients servers are configured. Discuss the various FTP and telnet commands.

What do you mean by application layer? When web pages are sent out, they are prefixed by MIME headers why? Explain with reasons. What do you mean by domain name system? How does it work? Explain resolution process. Explain about a typical resolution process in DNS.

* 10 5 1 0 1 *

PAPER ID: 101501

Time: **3 Hours]**

1.

a.

b.

c.

UNIT-I

[Total Marks: 100

(Do any Four)

$4 \times 5 = 20$

Draw the block diagram of a general communication model and explain the function of each block.

What are the different types of transmission technology? Explain different types of networks on the basis of transmission technology.

Describe the various technologies that can be used to form a LAN.

d.

Discuss the important design issues for various layers. Write a short notes on connection oriented and connectionless services.

e.

101501]

[2500]

101501]

Explain briefly the bus backbone and star backbone also state and explain different types of delays.

1

[Contd...

f.

2 a.

b.

C.

Distinguish between TCP/IP and OSI reference models. Which model is more popular and why?

UNIT-II

(Do any Two)

$10 \times 2 = 20$

State the various design issues for the data link layer. Sketch the Manchester and differential Manchester encoding for the bit stream 0001110101

Explain about role of minimum distance in error correction and detection. Calculate CRC for a 10 bit sequence 1010011110. The generator polynomial is $x^3 + x + 1$.

Discuss the role of protocols in LAN technology with examples also state advantages and disadvantages of FDDI and Ethernet LAN technologies.

3. a.

101501]

b.

What are the network design issues involved in designing a typical network and what are the supporting design tools available to make the design as a good design. Explain how these design tools help to address design issues.

C. An ATM network uses a token bucket scheme for traffic shaping. A new token is put into a bucket every 5 micro seconds. What is maximum sustainable net data rate excluding header bits?

4.

a.

b.

UNIT-IV

(Do any Two)

10×2=20

Explain about the communication taking place between the session layer, presentation layer and transport layer. Also explain the duties of presentation layer.

Illustrate about the TCP header and working of TCP protocol. Also explain about the various fields of TCP header with the help of neat diagram.

What do you mean by congestion control and QoS? Justify: presentation layer carries out the job of transition. What are the various security threats?

10×2=20

C.

UNIT-III:

(Do any Two)

A computer of 6 Mbps network is regulated by token bucket. The token bucket is filled at a rate of 1 Mbps. It is initially filled to capacity with 8 megabits. How long can be computer transmit at the full 6 Mbps.

[Contd...

101501]

3

[Contd...

PAPER ID: 9972

TCS-604

Printed Pages: 4 Paper ID and Roll No. to be filled in your Answer Book
Roll No.

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Xohay prove slit **B. Tech.**

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290 hay (SEM. VI) EXAMINATION, 2012

Time : **3 Hours]**

Note :

GRAPH THEORY

lib pitemmya b[Total Marks: **100**

Attempt all questions. All questions carry equal marks. Wera dqsig asingtlims bus
snolud ended

bris ad (ii)g State rank-nullity theorem. Define covering arqung seort and partitioning of a
graph.

(b) nie(i) to blf B is a circuit matrix of a connected graph G with e edge and n vertices,
prove that

nistre bos rank of $B - e - n + 1$. eniş

(5)

no 19 (ii) 280 Define cut-set matrix $C(G)$ and show that the rank of cut-set matrix is
equal to the atinio stporank of the incidence matrix.

(c)

Svo

The adjacency matrix of a graph G is given below. Using powers of this matrix determine
whether arisg owl vos tamen A G is connected or not. servisaid bns sont betoon site

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(c)

(d)

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(C)

(6)

1

(d)

Attempt any four parts: inay xe diy dast (a) Define geometric dual and combinatorial dual of ho ba graph. Give an example of a self dual graph. (b) Let G be a graph with n vertices which is isomorphic to its complement then show that o

$n - 1$ is divisible by 4.

Discuss the traveling salesman problem.

Construct a graph with vertex connectivity = 3, construct a graph with verte edge connectivity = 4 and minimum degree of a einshoqmoq batannos

vertex

5.

(e) Define a chromatic number. Find the chromatic polynomial for the cycle of length 4. Hence find

its chromatic number.

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DS-9972]

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LS [**2380**]

DS-99721

spanning

What is a spanning tree and a minimum spanning tree? Give an example for each.
What are types of enumeration? Explain.

3×4

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1

ES[**Contd...**

28 Attempt any two parts:

(a)(i)

M03-29T

=

Let G be a bipartite graph with 22 vertices with partite sets U and W , where $|U| = 12$. Suppose that every vertex in U has degree three, while every vertex of W has

STOS degree 2 or 4. How many vertices of G have degree 220.

(ii)

10×2

emit

Explain the operations of union, intersection and symmetric difference of two simple graphs with an example for each.

(b)

XE

Define Eulerian and Hamiltonian graph. Draw a graph with six vertices which

To sub(1)

which is

Hamiltonian and non-eulerian (ii) Eulerian

and non-hamiltonian (iii) Non-eulerian and non-

hamiltonian

(iv) Eulerian and Hamiltonian.

(c) Let G be a graph with vertices $n \geq 2$.

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(b) Draw the two Kuratowski's graphs and state the properties common to these graphs. Write a note on thickness and crossings in graphs.

design b(i)

ja svog

(c) (i)

Define the terms eccentricity and center in

half of a tree. Prove that every tree has either one

center or two centers.

t

(ii) Explain the concept of fundamental circuits in a directed graph.

would navigate in a direct

nadiadw enimmetab xitam air to enswog grial Attempt any two parts:

(a) (i)

ms Prove that if $\deg(v)$ for every

$>$

$n-2$

(b)

vertex v of G , then G contains at most two

connected components.

$2 = \text{xorov}$

(c)

(ii)

Explain the concept of 1-isomorphism and

brin sono

3

on

2-isomorphism with an example for each.

Tadium aitamos ati

Attempt any two parts insqa sai tadW

(a) Prove that if G is graph with vertices $n > 3$ and number of edges in such that $m > 3n-6$, then G is nonplanar.

DS-9972]

2

5

10×2

(ii)

(5)

ton, to belostno zi D Define a rooted tree and binary tree. Prove that a simple graph G is a tree if and only if there is a unique path between every pair of vertices.

Mention the difference between Prim's algorithm and Kruskal's algorithm for finding the minimum spanning tree with the help of an example.

Write Dijkstra algorithm for finding the shortest distance. Explain with the help of an example. Define distance, diameter, radius, spanning tree and minimum spanning tree for a graph.

Attempt any two parts:

(a) (i)

[Contd...

DS-9972]

Define incidence matrix of a graph. Prove that if $A(G)$ is an incidence matrix of a connected graph G with n vertices then rank of $A(G)$ is $n-1$.

3

[[Contd...

Describe three circumstances under which blocking

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WI/O should be used. Describe three circumstances under which non blocking I/O should be used. Why not just implement non blocking I/O and have processes busy-wait until their device is

DS-1077]

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PAPER ID: 1077

TCS-601

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Paper ID and Roll No. to be filled in your Answer Book Roll No. Insminitab edi bris dio habiano? P

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(d)

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(SEM. VI) EXAMINATION, 2012

OPERATING SYSTEM

Time: **3 Hours]** 1905-19oubor

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ed [Total Marks: **100 (5) 5×4=20**

Attempt any four out of six, yno ewolls 219Hud is wolls of minogle odi viiboM .amir

(a)

List four steps needed to run a program on a completely dedicated machine.

(b) What are the main differences between novoperating systems for mainframe computers Jabal and PCs ? vs

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(c) Define the essential properties of the following types of operating systems
bnuod U40.evista yliamsmaq Jon **Jay** (i) Distributed

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ms 29550 UD A (d)ub What are the five major activities of an bolub operating system in regard to process aslubomanagement ?m wod

(ii) Parallel mogls "gniluberoz (3)

ΑΠΟ ΠΟ

(e) What are the three major activities of an olqislu operating system in regard to secondary-storage management ? hooned say

(f) What is the main advantage for an operating- Sosystem designer of using a virtual-machine

architecture?

sqy

[**2380-20**

DS-1077]

1

[**Contd...**

2

3

Attempt any two out of three:

10a-837 10×**2=20**

(a) Describe the differences among short-term, medium-term, and long-term scheduling.

(b) What are the benefits and the detriments of each of the following? Consider both the systems and the programmers' levels.

(i) MAutomatic and explicit buffering

(ii) Send by copy and send by reference

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(c) The correct producer-consumer algorithm must allow only $n-1$ buffers to be full at any one time. Modify the algorithm to allow all buffers to be utilized fully. 29 o pai (6)

no te

Attempt any two out of three: 10×2=20 (a) Suppose that a scheduling algorithm favours those processes that have used the least processor time in the recent past. Why will this algorithm favour I/O-bound programs and yet not permanently starve CPU-bound programs?

as

(b) A CPU scheduling algorithm determines an order for the execution of its scheduled 22900 processes. Given n processes to be scheduled

on one processor, how many different schedules are possible? Give a formula in terms of n . (c) Show through an example that the multiple resource type banker's scheme cannot be implemented by individual application of the single-resource type scheme to each resource

DS-1077]

type.

2

97d1o9tidots

4

Attempt any two out of three: 10×2=20 (a) Why are page sizes always powers of 2? Why are segmentation and paging sometimes combined into one scheme? 120jton vW

(b)

Consider a paging system with the page table stored in memory.

(i)

(c)

If a memory reference takes 200 nano seconds, how long does a paged memory reference take?

(ii) If we add TLBs, and 75% of all page-table references are found in the TLBs, what is the effective memory reference time?

Suppose that your replacement policy in a paged system is to examine each page regularly and to discard that page if it has not been used since the last examination. What would you gain and what would you lose by using this policy rather than LRU.

5

Attempt any two out of three:

(a)

(b)

[Contd...

DS-1077]

10×2=20

What are the advantages and disadvantages of recording the name of the creating program with the file's attributes.

Consider a system that supports the strategies of contiguous, linked, and indexed allocation. What criteria should be used in deciding which strategy is best utilized for a particular file?

3

[Contd...

* 1 0 7 5

*

PAPER ID: 1075

TCS-503

Printed Pages: 4

Paper ID and Roll No. **to be** filled in **your Answer Book**

Roll **No.**

B. Tech.

(SEM. V) (ODD SEM.) EXAMINATION, 2012-13

DESIGN ANALYSIS OF ALGORITHMS

Time: **3 Hours]**

SECTION-1

[Total Marks: 100

(Attempt any four; each question carries 5 marks)

1.

$I_{s,n+1}$

$O(2n)$? Is $22n O(2)$?

2.

3.

Arrange the following in order of best to worst efficiency: $O(n^2 \log n)$, $O(n^2)$, $O(n^3)$, $O(n^2 \log \sqrt{n})$.

Give an analysis of worst case running time of quicksort. When does it occur?

4.

5.

6.

1075]

Give an array of 8 elements which is the worst case for

Insertion Sort.

Define the asymptotic notations used for best case average case and worst case analysis of algorithms.

Show that the solution $T(n) = T(n/2) + 1$ is $O(\lg n)$.

1

[Contd...

SECTION-2

(Attempt any two; each question carries 10 marks)

7.

8.

9.

10.

(A) Insert the following values one at a time into an initially empty red-black tree

41, 38, 31, 12, 19, 8.

(B) Show that, for arbitrarily large values of n , there are red-black trees with n nodes that have height $2 \lg n - O(1)$.

Suppose that a node x is inserted into a red-black tree with RB-INSERT and then immediately deleted with RB-DELETE. Is the resulting red-black tree the same as the initial red-black tree?

Insert the following nodes into an empty AVL tree. Show each step and what rotations are needed. Nodes to insert: 10, 40, 35, 25, 60, 30, 80, 50, 27, 28, 38

SECTION-3

(Attempt any two; each question carries 10 marks)

(A) What is the difference between a pure recursive solution and a dynamic programming solution to a problem? (B) What do we understand by the "optimal substructure property" in a dynamic programming problem?

11. Explain N-queens problem with an algorithm. Explain why backtracking is defined as a default procedure of last resort for solving problems.

12. (A) (B)

State if backtracking always produces optimal solution. Describe how to implement a queue using two stacks and $O(1)$ additional memory, so that the amortized time for any

enqueue or dequeue operation is $O(1)$. The only access you have to the stacks is through the standard subroutines PUSH and POP.

SECTION-4

(Attempt any two; each question carries 10 marks)

13. Solve the following instance of the single-source shortest paths problem with vertex "v3" as the source.

2

V1

V2

JO

2

8

2

90

Write the algorithm for the above problem.

6

14. Draw a (simple) directed weighted graph G with 10 vertices and 18 edges, such that G contains a minimum-weight cycle with at least 4 edges. Show that the Bellman-Ford algorithm will find this cycle.

15.

Show how to modify Dijkstra's algorithm for the case when the graph is directed and we want to compute shortest directed paths from the source vertex to all the other vertices.

1075]

2

[**Contd...**

1075]

3.

[**Contd...**

SECTION-5

(Attempt any two; each question carries 10 marks) 16. (A) How do we establish that a problem is NP-Complete? Given a graph G and vertices s and t , does G contain

(B)

a walk from s to t such each vertex of G is visited

at least once and at most 100 times.

17. What happens when we use the Goemans-Williamson algorithm for MAX-CUT on a triangle? Square? Pentagon? Specifically, what can be inferred on the algorithm's ratio and on the integrality ratio of the relaxation?

18. Show that the Next-Fit algorithm for BP (BIN-PACKING) achieves an asymptotic ratio of 2. Show that $FF(I) \leq NF(I)$.

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TCS-503

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Odd Semester Examination, 2017-18

B.TECH. (SEMESTER-V)

DESIGN & ANALYSIS OF ALGORITHM

Time: 03:00 Hours

Max Marks: 100

Note: Attempt **all** questions:

1.

Attempt **any four** questions:

(a)

What is B and B+ tree? Write four differences between them.

(b)

(c)

(d)

(e)

[5x4=20]

What is minimum spanning tree? Write two difference between prism and kruskal.

Priority Queue using Heap Data Structure has always been a data structure of choice for many applications. **Discuss its** beauty in terms **of its** salient features. How it performs in terms of access and modification. How it performs better than other peer formations in terms of resource complexity.

Differentiate between Greedy and Dynamic Algorithms with appropriate examples?

Insert all the elements in red black tree and show all the steps of insertion:

42,24,75,14,65,35,90,10,20,18,55,70,80,60,40

Attempt **any four** questions:

2.

O

(a)

(1)

[5x4=20]

Insert the following elements in B Tree (Order-3) and show **all** the steps of insertion.

42, 24, 75, 14, 65, 35, 90, 10, 20, 15, 55, 70, 80, 60, 40.

(2)

Delete 24 and 75. **Show all** the steps.

(b)

Explain disjoint **set** Operations Also Explain **the** applications of Disjoint -Set data Structures?

(c)

(d)

TCS-503/1680

Write algorithm for union **of two** binomial heaps. Also explain the properties of Binomial **trees?**

Let $A = \{a/20, b/15, c/5, d/15, e/45\}$ **be** the alphabet and its frequency distribution.

Construct the Huffman tree and algorithm.

1

[P.T.O.]

(e)

Explain the process of deletion in red black tree.

3.

Attempt **any two** questions:

[10x2=20]

(a) Write algorithm **for** n queen problem using Backtracking approach. Discuss your algorithm using 4x4 chess board.

(b)

(c)

Define complete KMP algorithm for string matching. It should give the preprocessing and searching algorithms separately. Given the pattern: 101001 and string: 110101010100111111, solve this using KMP.

Write algorithm for 0-1 knapsack problem using dynamic programming. Apply the **same** to find optimal items in the Knapsack in the following example and show all the intermediate steps. Consider 4 items along their respective weights and values

Item i Value (v_i) Weight (w_i)

1 15 1

2 10 5

3 9 3

4 54

The maximum capacity **of the** knapsack is $W = 8$ Fill **the** knapsack such that knapsack should **not exceed its** maximum capacity and **it** should have maximum profit value.

4.

Attempt **any two** questions:

[10x2=20]

(a) What are approximation algorithms? Explain traveling salesman problem with example. Also give 2 differences between binomial and Fibonacci heaps.

(b)

TCS-503/1680

31

S

2

E

A

3

1

G

4

C

3

5

T

B

F

2

(c)

5.

Consider the following directed graph. There are a multiple shortest paths between 12M vertices S and T. Which one will **be** reported by Dijkstra's shortest path algorithm? Write the sequence of vertices and cost of the shortest path from S to T. Assume that, in any iteration the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.

What is Topological sorting? Write complete algorithm for Topological Sort.

Attempt **any two** questions:

[10x2=20]

(a)

(1)

Compare P and NP.

(2)

Differentiate among Divide and conquer, Dynamic Programming and backtracking.

(3)

Define Queue and its possible operations.

(4)

Differentiate between BFS and DFS.

(5)

Discuss memory representations of a graph

(b)

Given two string sequences, write an algorithm to find the longest common (10) subsequence and its length using dynamic programming. Apply the algorithm to following two sequences: $X = \langle A, B, C, B, A \rangle$ and $Y = \langle B, D, C, A, B \rangle$.

(c)

Write short notes on:

i.

Genetic Algorithms

ii.

Intelligent Algorithms

iii.

Parallel Algorithms

iv.

NP-complete problem

TCS-503/1680

3

Sub Code: BCST603 ROLL NO.....

EVEN SEMESTER EXAMINATION, 2023 – 24

3rd yr. B.Tech. – Computer Science & Engineering

DATA ANALYTICS

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a)What are the three modes that Hadoop can Run? b) How will you carry out Independence of Attributes test? c) What is Big Data? Explain 5V model in Big Data? d) Differentiate between Null and Alternative Hypotheses? e) Write the test procedure for testing two proportions in the case of large samples. f) What are the two types of errors in testing of hypothesis?	5x4=20
Q 2.	Answer any four parts of the following. a) Describe the following in terms of HDFS: i) Name Node and Data Node ii) Blocks and Input Splits iii) Hadoop Streaming iv) Hadoop Pipes v) Data Locality Optimization b) Summarize the various types of crowd sourcing involved in analytics? c) In a two-way ANOVA, the number of factor A levels is 4, the number of factor B levels is 6, and the number of replicates is 2. Find the number of degrees of freedom for interaction? d) What are data processing operations in PIG? e) Describe the difference between one-way ANOVA and two-way ANOVA. Provide an example of when ANOVA would be used? f) Why do we need Non-Parametric tests? Briefly explain the assumptions made in the case of Non-Parametric tests?	5x4=20

Q 3.	Answer any two parts of the following. a) What is Hadoop? Draw diagram to discuss features of Hadoop. How do you analyze the data in Hadoop? b) Explain the process of installing HIVE & features of HIVE? c) Explain Grouping, Join, CoGroup, Cross, & Group in data?	10x2= 20
Q 4.	Answer any two parts of the following. a) Define HDFS? Describe namenode, datanode and block. Explain HDFS operations in detail? b) What is Zookeeper explain its features with applications? c) Discuss about How E-Commerce is Using Big Data to Improve Business in detail?	10x2= 20
Q 5.	Answer any two parts of the following. a) What is PIG? Explain its installing process and two execution types or modes in PIG? b) Describe map reduce framework in detail? Explain how map reduce jobs run on YARN? Draw architectural diagram for physical organization of nodes? c) What is open-source technology? Explain differences of inter and trans firewall analytics?	10x2= 20

**Sub Code: CST-011 ROLL NO..... ODD SEMESTER EXAMINATION,
2024 – 25**

II Year (III Sem) B.Tech.

**CS&E/CS&E (Data Science)/CS&E (AI&ML)/ CS&E (IOT & Cyber Security)/CS&IT
DATABASE MANAGEMENT SYSTEM**

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) Describe DBA. What are the roles of DBA (5 marks) (ii) Define the role of DDL in a database management system. Provide examples of common DDL commands and their usage. (5 marks) b) Explain the three-level database architecture (external, conceptual, and internal levels). Discuss how this architecture supports data abstraction and independence. (10 marks) c) Define the Entity-Relationship (ER) model and its components. Create an ER diagram for a university database that includes entities, attributes, relationships, and cardinality. (10 marks)
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) What is the Data Model? Explain Data independence with a suitable diagram. (5 marks) (ii) Explain the Structure of Relational Databases with examples. (5 marks) b) Consider the following schema for institute library: (10 marks) Student (RollNo, Name, Father_Name, Branch) Book (ISBN, Title, Author, Publisher) Issue (RollNo, ISBN, Date-of-Issue) Write the following queries in SQL and relational algebra: · List roll number and name of all students of the branch 'CSE'. · Find the name of a student who has issued a book published by 'ABC' publisher. · List title of all books and their authors issued to a student 'RAM'. · List title of all books issued on or before December 1, 2020. · List all books published by publisher 'ABC' c) What are Joins? Discuss all types of Joins with the help of suitable examples. (10 marks)
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) Define functional dependencies in the context of relational databases. Explain their significance in database design with examples. (5 marks) (ii) Explain the concept of decomposition in relational database design. Discuss the significance of decomposition in achieving a well-designed schema. (5 marks) b) Given a relation R (A, B, C, D) and Functional Dependency set FD = {AB → CD, B → C}, determine whether the given R is in 2NF? If not, convert it into 2 NF (10 marks) c) Define Third Normal Form (3NF). Discuss the differences between 2NF and 3NF with examples. Why is 3NF important in database design? (10 marks)
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) Define ACID properties in the context of database systems with examples. (5 marks)

	(ii) Describe the Two-Phase Commit protocol in detail, including the roles of the coordinator and participants. Explain the steps involved and the states of the protocol. (5 marks) b) Compare and contrast Locking Protocols and Timestamp-Based Protocols in concurrency control. Provide examples to illustrate their differences. (10 marks) c) Explain the benefits and potential challenges of implementing save points in a distributed database environment. (10 marks)
Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) Explain the various file organization techniques with examples. (5 marks) (ii) What are ordered indices, and why are they important in databases? (5 marks) b) Explain the steps involved in query processing in a database management system. Provide an example of how a query is processed and optimized. (10 marks) c) Define data warehousing and its significance in decision-making. Discuss the various components of a data warehouse architecture (10 marks)

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Odd Semester Examination 2018-19

B.TECH. (CSE) (SEMESTER-V)

DESIGN & ANALYSIS OF ALGORITHM

Time: 03:00 Hours

bns

Note: Attempt **all** sections.

Section A

1.0sic

Note: Attempt **any four**. All question carry equal marks:

Define the following terms:

(a) NP complete problem

(b)

Satisfiability problem

(c)

Clique Problem

(d) Breadth First Search

(e)

Depth first Search

J noissa

Section B

Max Marks: 100

[4×5=20]

Note: Attempt **any four**. All question carry equal marks:

[4×5=20]

1.

What do you understand by these two programming paradigm i.e. Greedy approach and Dynamic programming?

N

2.1

What do you understand by minimum spanning tree? Is it possible for a graph to have multiple spanning trees? If yes, provide a suitable example.

3. Give bellman-ford algorithm for single source shortest path.

TCS-503/2280

(1)-(9)

03 [P.T.O.]

4.

What do you understand by minimum spanning tree? Is it possible for a graph to have multiple spanning trees? If yes, provide a suitable example.

5.

Every tree is a graph but every graph is not a tree. What do you understand by this statement? Please explain using a suitable example

Section C Section

Note: Attempt **any four**. All question carry equal marks:

1.

[4x5=20]

Differentiate between Min-Heap and Max-Heap? Apply and show stepwise implementation

2.

Differentiate between sparse and dense graphs? Which representation of graph is better and in which case?

3.

4.

Explain how we can solve a given 0/1 Knapsack problem with the help of Branch and Bound techniques

Give algorithm to solve the 8-Queen problem using backtracking approach.

5.

Give algorithm of Heap Sort

Section D

Note: Attempt **any two**. All question carry equal marks:

[2×10=20]

1.

Discuss time complexity of Insertion Sort Algorithm in its best case.

2.

Apply and show stepwise implementation of Max-Heapify (A,2) on array A = {16, 4, 10, 14, 7, 9, 3, 2, 8, 1}, where array index starts at 1.

DHA 13.0149 Briefly explain the difference between fractional and 0/1 knapsack problem. Write algorithm of fractional knapsack and apply it on following set of weights and their profits to maximize the profit where Knapsack capacity is 60. Weights {40, 10, 20, 24} and Profits = {280, 100, 120, 120}.

TCS-503/2280

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Section E

Note: Attempt **any two**. All question carry equal marks:

1.

2.

3.

[2×10=20]

Write pseudocode and recurrence relation for the recursive function that prints Nth Fibonacci number. Solve the recurrence relation to get time complexity of the algorithm. What will be the space complexity of this program and why?

Why do we use Huffman codes? A data file of 100,000 characters contains only the characters a-f, with the frequencies (in thousands) indicated as follows - {a:45, b:13, c:12, d:16, e:9, f:5}. If we assign each character a 3-bit codeword, we can encode the file in 300,000 bits. Your task is to assign variable-length code and encode the file in

optimal number of characters. How much space are you able to save with variable At length encoding?

Write a recurrence relation when quick sort repeatedly divides the array in to two parts of 90% and 10%. Derive the time complexity in this case. Show the recursion tree while deriving time complexity and find the difference in heights of both the extremes of recurrence tree. What do you understand by this analysis?

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B

Note A

TCS-503/2280

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Odd Semester Examination 2018-19

B.TECH. (CSE) (SEMESTER-V)

DESIGN & ANALYSIS OF ALGORITHM

Time: 03:00 Hours

bns

Note: Attempt **all** sections.

Section A

1.0sic

Note: Attempt **any four**. All question carry equal marks:

Define the following terms:

(a) NP complete problem

(b)

Satisfiability problem

(c)

Clique Problem

(d) Breadth First Search

(e)

Depth first Search

J noissa

Section B

Max Marks: 100

[4×5=20]

Note: Attempt **any four**. All question carry equal marks:

[4×5=20]

1.

What do you understand by these two programming paradigm i.e. Greedy approach and Dynamic programming?

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What do you understand by minimum spanning tree? Is it possible for a graph to have multiple spanning trees? If yes, provide a suitable example.

3. Give bellman-ford algorithm for single source shortest path.

TCS-503/2280

(1)-(9)

03 [P.T.O.]

4.

What do you understand by minimum spanning tree? Is it possible for a graph to have multiple spanning trees? If yes, provide a suitable example.

5.

Every tree is a graph but every graph is not a tree. What do you understand by this statement? Please explain using a suitable example

Section C Section

Note: Attempt **any four**. All question carry equal marks:

1.

[4x5=20]

Differentiate between Min-Heap and Max-Heap? Apply and show stepwise implementation

2.

Differentiate between sparse and dense graphs? Which representation of graph is better and in which case?

3.

4.

Explain how we can solve a given 0/1 Knapsack problem with the help of Branch and Bound techniques

Give algorithm to solve the 8-Queen problem using backtracking approach.

5.

Give algorithm of Heap Sort

Section D

Note: Attempt **any two**. All question carry equal marks:

[2×10=20]

1.

Discuss time complexity of Insertion Sort Algorithm in its best case.

2.

Apply and show stepwise implementation of Max-Heapify (A,2) on array A = {16, 4, 10, 14, 7, 9, 3, 2, 8, 1}, where array index starts at 1.

DHA 13.0149 Briefly explain the difference between fractional and 0/1 knapsack problem. Write algorithm of fractional knapsack and apply it on following set of weights and their profits to maximize the profit where Knapsack capacity is 60. Weights {40, 10, 20, 24} and Profits = {280, 100, 120, 120}.

TCS-503/2280

(2)

Section E

Note: Attempt **any two**. All question carry equal marks:

1.

2.

3.

[2×10=20]

Write pseudocode and recurrence relation for the recursive function that prints Nth Fibonacci number. Solve the recurrence relation to get time complexity of the algorithm. What will be the space complexity of this program and why?

Why do we use Huffman codes? A data file of 100,000 characters contains only the characters a-f, with the frequencies (in thousands) indicated as follows - {a:45, b:13, c:12, d:16, e:9, f:5}. If we assign each character a 3-bit codeword, we can encode the file in 300,000 bits. Your task is to assign variable-length code and encode the file in

optimal number of characters. How much space are you able to save with variable At length encoding?

Write a recurrence relation when quick sort repeatedly divides the array in to two parts of 90% and 10%. Derive the time complexity in this case. Show the recursion tree while deriving time complexity and find the difference in heights of both the extremes of recurrence tree. What do you understand by this analysis?

X

B

Note A

TCS-503/2280

(3)

TCS 503 Roll No.

Odd Semester Examination, 2019-20 B.Tech (CSE/IT, 5th Semester) Design & Analysis of Algorithms

Max. Marks: 100 Total no. of printed pages: 2

(5 × 4 = 20)

Use recursion tree to solve the following recursive equation also prove it correct using substitution method.

Write the algorithm for Max- Heapify procedure. Illustrate the functioning of Heap Sort on the following array: A= <55,62,41, 27,15,20,5,60>

Write a pseudocode for multiplying two matrices of size n *n also derive the running time complexity of your algorithm?

Time: 3 Hours

Note: (i)

Attempt **ALL** questions.

(ii) Qu(iii)

All Questions carry equal marks.

Be precise in your answer.

Q1.

Attempt **any four** parts of the following:

(a)

Explain Heap sort with suitable examples.

(b)

$$T(n) = 4T(n/3) + n$$

(c)

(d)

(e)

(f)

Explain Master's Theorem.

Q2.

(a)

(b)

b) O (Big-Oh?)

(c)

(d)

(e)

Solve the following recurrence relations

$$T(n) = T(n-1) + T(n-2) \text{ for } n > 1 \text{ and } T(0)=0, T(1)=1$$

$$T(n) = n + T(n/10) + T(9n/10)$$

Attempt **any four** parts of the following:

$$(5 \times 4 = 20)$$

Prove that a red-black tree with n internal nodes has height almost $2\log(n+1)$? Define following asymptotic notations

a) $\rightarrow$ (Theta)

Write quick sort algorithm and discuss its worst case and best case complexity. Apply quick sort algorithm on array $A = \langle 76, 22, 50, 79, 98, \mathbf{87}, 16, 3, 25, 100 \rangle$ Explain searching operation in B-tree.

What is a B-Tree? Draw a B-tree of order 3 inserting the following elements in sequence

92 19 6 7 11 8 22 4 5 16 36 20 78?

(f) Explain Data Structure for Disjoint Sets.

Attempt **any two** parts of the following:

(10 $\times$ 2 = 20) Discuss the dynamic programming solution to largest common subsequence (LCS) problem. Write an algorithm to compute an LCS of two given strings. What is 0/1 knapsack problem? Does greedy method effective to solve the 0/1 knapsack problem?

Q3.

(a)

(b)

(c)

When and how dynamic programming approach is applicable? Discuss the matrix chain multiplication with respect to dynamic programming technique.

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Attempt **any two** parts of the following:

Q4.

(a)

(b)

(c)

Q5.

(a)

(b)

(c)

(10×2=20) Explain prims algorithm to find the minimum cost spanning tree of an n-vertex undirected network with example.

Write the algorithm for Bellman Ford algorithm. Also find the shortest path using the same.

Explain greedy algorithms with the help of suitable examples?

Attempt **any two** parts of the following:

(10 × 2 = 20)

Explain backtracking why it is required? Also explain B trees

What is string matching algorithm? Explain Boyer-Moore algorithm for string matching for: abcaabccaabbabcaacacbbbss

Compute worst time complexity of this algorithm? Explain in detail travelling salesman problem?

2

Sub Code: BCST 503 ROLL NO..... ODD SEMESTER EXAMINATION,
2024 – 25

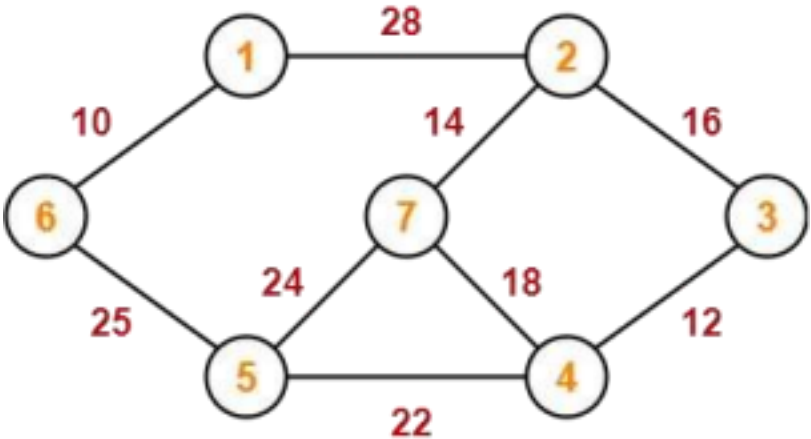
III Year (V Sem) B.Tech.: CS&E/IT

Design and Analysis of Algorithms

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) Define algorithm and its performance analysis. (5 marks) (ii) Explain the concept of randomized algorithm with a suitable example (5 marks) b) Explain the working of merge sort algorithm. Assume that a merge sort algorithm in the worst case takes 30 seconds for an input size 64. Describe the maximum input size (approximated) of a problem that can be solved in 6 minutes. (10 marks) c) Explain the worst and best case time complexity of Quick sort. Consider the Quick Sort algorithm with the last element chosen as the pivot. If the goal is to sort the given $a=[30,40,50,60,70,80]$ array $a=[30,40,50,60,70,80]$ in ascending order, how many swaps will occur during the execution of the algorithm? (10 marks)
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) Describe the algorithms of union and find operations. (5 marks) (ii) Describe the ways to determine Hamiltonian cycle in a graph. (5 marks) b) Define graph coloring problem. Describe the chromatic number required using graph coloring technique of the given connected graph. (10 <pre>graph TD e --- d e --- f e --- g d --- c d --- g f --- g g --- c g --- a g --- b f --- a a --- b b --- c</pre> marks) c) Define backtracking technique of algorithm. Describe the concept of N-Queen problem with a suitable example. (10 marks)

Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) Define minimum cost spanning tree with a suitable example. (5 marks) (ii) Describe the edge relaxation in dijkstra algorithm. (5 marks) b) Consider the jobs, their deadlines and associated profits as shown in table Jobs J1 J2 J3 J4 J5 J6 Deadlines 5 3 3 2 4 2 Profits 200 180 190 300 120 100
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	Describe the following : i. Optimal schedule to obtain the maximum profit. ii. Maximum earned profit. (10 marks) c) Describe the order of vertices of minimum spanning tree (MST) for the given graph using Prim's and Kruskal algorithm. (10 marks) 
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) Differentiate between greedy and dynamic programming approach in algorithms. (5 marks) (ii) Describe the working of all pairs shortest path problem. (5 marks) b) Define optimal binary search tree with recurrence relation. Consider the keys and frequency as mentioned below: Keys 10 20 30 40 Frequency 4 2 6 3 Describe optimal binary search tree (BST) using dynamic programming approach. (10 marks) c) Describe chained matrix multiplication recurrence relation. Consider A1, A2, A3 and A4 be four matrices of dimensions 10×5, 5×20, 20×10 and 10×5, respectively. Describe the minimum number of scalar multiplications

	required to find the product $A_1A_2A_3A_4$ using the chained matrix multiplication method. (10 marks)
Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) Define Cook's theorem. (5 marks) (ii) Describe computability classes of problems. (5 marks) b) Explain LC and FIFO Branch and Bound techniques with a suitable example. (10 marks) c) Consider the profits and weight (in kgs) of the four objects as mentioned in table: Objects ob1 ob2 ob3 ob4 Profit 10 10 12 18 Weight 2 4 6 9 Describe the order of weights to be used in 0/1 knapsack problem using branch and bound technique. (10 marks)
